ASSIGNMENT – 1

CRYPTOGRAPHY

EVALUATE, DESIGN AND DEFEND A SECURE PATIENT-RECORD EXCHANGE STRATEGY FOR A MULTI-HOSPITAL HEALTHCARE NETWORK

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Scenario and Objective

A regional healthcare network contains one central hospital and 25 satellite clinics. Clinics exchange patient records, diagnostic reports, prescriptions and billing data with the central hospital over a public network. Users have different security clearances: Medical Director – Top Secret; Specialist Doctors – Secret; Nurses and Lab Staff – Confidential; Billing and Front-Desk Staff – Restricted; Patients – Public.

The objective is to evaluate classical encryption and access-control techniques, demonstrate their mathematical properties and limitations, and recommend a practical design. Classical ciphers are used here for academic evaluation only. They are not safe for real patient records.

Assumptions and notation

- Letters are represented as A=0, B=1, …, Z=25 for mod-26 examples.
- For the extended Hill cipher, the alphabet is A–Z followed by 0–9, giving 36 symbols. Separators such as spaces and hyphens are preserved and are not encrypted.
- Cipher examples use uppercase text and remove spaces where the algorithm requires a fixed alphabet.
- Risk scores use Likelihood × Impact, each rated from 1 (low) to 5 (very high).
- All example patient details are fictional.

TASK 1 – Evaluate Threats, Attacks and Risks (4 Marks)

A threat is a possible cause of harm; a vulnerability is a weakness that can be exploited; an attack is the action used to exploit it; and risk is the likelihood and impact of the resulting harm.

Term	Healthcare example
Threat	A malicious insider or external attacker wants to steal or alter patient records.
Vulnerability	The same classical key is reused at many clinics and staff share it.
Attack	An attacker copies ciphertexts, performs frequency analysis, or changes a prescription.
Risk	A forged dosage may cause patient injury; likelihood and impact determine its priority.

Attack	Type	Likelihood	Impact	Score	Reason / priority
Prescription or lab-value modification	Active	4	5	20 – Critical	Can directly cause wrong treatment; address first.
Masquerade as a doctor	Active	4	5	20 – Critical	False orders may be accepted if identity is not authenticated.
Replay of an old prescription	Active	4	5	20 – Critical	An expired or duplicate order may be administered.
Ciphertext eavesdropping	Passive	5	3	15 – High	Public network makes copying easy; weak ciphers expose diagnoses.
Traffic analysis	Passive	4	3	12 – High	Message timing and size can reveal clinic activity.
Denial of service	Active	3	4	12 – High	Clinicians may lose access during urgent treatment.

Priority decision: prescription modification, doctor masquerade and replay are addressed first because they can create immediate patient-safety harm. Confidentiality is important, but an altered or unauthenticated order can be more dangerous than disclosure alone.

TASK 2 – Map and Judge Security Goals (3 Marks)

Function	Most important goals	Reason
Record upload	Confidentiality, integrity, authentication, availability	Records must be private, unaltered, sent by an approved clinic and available when needed.
Prescription transmission	Integrity, authentication, non-repudiation, availability, confidentiality	Correct patient, medicine and dosage are safety-critical; sender identity must be established.
Specialist consultation	Confidentiality, authentication, integrity, availability	Only authorized clinicians should see the case and advice must remain unchanged.

Classical substitution and transposition ciphers mainly provide concealment (weak confidentiality). They do not independently provide authentication, integrity, availability or non-repudiation. A correctly decrypted message only shows that the receiver used a compatible key. It does not prove who sent it: anyone possessing the shared key can create a valid-looking ciphertext. Nor does successful decryption prove that the dosage was not changed. A separate integrity check and authenticated sender mechanism would be required in a real system.

TASK 3 – Apply and Critique the Bell–LaPadula Model (5 Marks)

Bell–LaPadula (BLP) is a confidentiality model. Its simple security property is “no read up”: a subject cannot read an object at a higher classification. Its *-property is “no write down”: a subject cannot write information to a lower classification.

Level	Assigned users / data
Top Secret (TS)	Medical Director; highly sensitive strategic records.
Secret (S)	Specialist doctors; specialist diagnosis and consultation records.
Confidential (C)	Nurses and lab staff; nursing notes and lab records.
Restricted (R)	Billing and front-desk staff; billing and appointment records.
Public (P)	Patients; approved patient-facing information.

Request	Decision	Reason
Medical Director (TS) reads Secret diagnosis	Allow	Read down is permitted.
Specialist (S) reads Confidential lab result	Allow	Read down is permitted.
Nurse (C) reads Secret allergy record	Deny	Read up violates simple security property.
Billing clerk (R) reads Confidential diagnosis	Deny	Read up violates simple security property.
Nurse (C) writes a note to Restricted billing file	Deny	Write down violates *-property.
Specialist (S) writes to Secret consultation record	Allow	Same-level write is permitted.
Billing clerk (R) writes to Restricted invoice	Allow	Same-level write is permitted.

Medical-setting critique

- Emergency access: a nurse who needs a Secret allergy record is denied by ordinary BLP rules. A controlled break-glass procedure may temporarily authorize access, but it must record the user, reason, patient, time and data viewed, and trigger later review.
- Doctor writing a billing note: a Secret-level doctor cannot directly write sensitive information into a lower Restricted file under no-write-down. A trusted declassification process should extract only necessary billing facts, such as procedure code and approved amount, without copying diagnosis details.
- Tranquility: changing a subject's or object's classification during operation can create security problems. Strict tranquility avoids changes; a controlled version permits carefully authorized changes.
- Trusted subjects: specially authorized processes may perform declassification or emergency actions, but they must be narrowly scoped and audited.
- Limitations: BLP focuses on confidentiality, not correctness of records. It does not itself stop prescription tampering, covert channels, or denial of service. Its rigidity can obstruct emergency care.

Verdict: BLP is useful as a confidentiality policy foundation, but it is not sufficient alone for healthcare. It must be combined with integrity checks, authenticated identities, audit logs, emergency procedures and availability controls.

TASK 4 – Design the Cryptographic Strategies (4 Marks)

Three different classical approaches are demonstrated below. The examples are for learning, not deployment.

A. Caesar cipher

Key: integer k from 0 to 25. Encryption: $C = (P + k) \bmod 26$. Decryption: $P = (C - k) \bmod 26$.

Example: plaintext PATIENTIDDIAGNOSIS, key k=3. Each letter is shifted three places. Ciphertext: SDWLHQWL GGLDJQRVLV (spaces shown only to separate the illustrative groups; actual output is SDWLHQWLGGLDJQRVLV).

B. Vigenère cipher

Key: a repeating keyword. Convert letters to numbers 0–25. Encryption: $C_i = (P_i + K_i) \bmod 26$. Decryption: $P_i = (C_i - K_i) \bmod 26$.

Example: plaintext PATIENTDIAGNOSIS, key CARE repeated to match length. The key stream is CARECARECARECARE; ciphertext is RAKMGNKHKAXRQSZW. This uses multiple Caesar shifts and hides single-letter frequencies better than Caesar when the key is sufficiently long and not repeated predictably.

C. Hill cipher (2 × 2 demonstration)

Key matrix K = [[3,3],[2,5]], with determinant 9. Since gcd(9,26)=1, the matrix is invertible modulo 26. Encryption uses C = KP mod 26 for two-letter column vectors.

The modular inverse of 9 mod 26 is 3 because $9 \times 3 = 27 \equiv 1 \pmod{26}$. The inverse matrix is $K^{-1} = 3 \times [[5,-3],[-2,3]] \bmod 26 = [[15,17],[20,9]]$.

Example: plaintext HI gives vector $[7,8]^T$. $K[7,8]^T = [45,54]^T \equiv [19,2]^T \bmod 26$, which is TC. Decrypting TC with K^{-1} returns HI.

D. Columnar transposition

Write plaintext row-wise under a numbered keyword and read columns in key order. This rearranges positions but does not change letter counts.

Example keyword 3-1-4-2 and plaintext PATIENTRECORDS. Pad with X to 16 characters and write four columns: P A T I / E N T R / E C O R / D S X X. Read columns in key order 1,2,3,4 (the columns labelled 1,2,3,4 are respectively the second, fourth, first and third columns). Ciphertext: ANCSIRXXPEEDTTOX.

Selection justification: Caesar is a simple baseline; Vigenère demonstrates polyalphabetic substitution; Hill demonstrates matrix-based block substitution and linear algebra; columnar transposition demonstrates position rearrangement. Together they expose different strengths and weaknesses.

TASK 5 – Mathematical Analysis of the Schemes (5 Marks)

Scheme	Key-space size	Encryption function / observation
Caesar	26 shifts, or 25 non-trivial keys	$C=(P+k) \bmod 26$; trivial to brute-force.
Monoalphabetic substitution	$26! \approx 4.03\times10^{26}$ permutations	A large key space, but each plaintext letter always maps to the same ciphertext letter.
Vigenère, key length m	26^m possible keys (before excluding weak keys)	$C_i=(P_i+K_i) \bmod 26$; repeating keys create periodic patterns.
Columnar transposition, n columns	n! column permutations	Positions change, but letter frequencies remain identical.
Hill, n×n over mod 26	Number of invertible matrices; for 2×2, not simply 26^4	$C=KP \bmod 26$. K^{-1} exists exactly when $\gcd(\det K,26)=1$.

Hill invertibility condition: a matrix has an inverse modulo 26 if and only if its determinant is relatively prime to 26. This means the determinant must be odd and not divisible by 13. For the 2×2 key above, $\det(K)=3\times5-3\times2=9$ and $\gcd(9,26)=1$, so the inverse exists.

Why a large key space is not enough: monoalphabetic substitution has 26! possible keys, but it preserves single-letter frequency and common patterns. In English, E, T, A and common digrams occur frequently. An attacker can use frequency and pattern analysis rather than trying every key. Security depends on resistance to attacks, not key count alone.

Extended Hill requirement: use alphabet A–Z,0–9 with values 0–35 and arithmetic mod 36. One simple invertible 3×3 key is K=[[1,2,0],[0,1,0],[0,0,1]]. Its determinant is 1, so gcd(1,36)=1. Its inverse is [[1,34,0],[0,1,0],[0,0,1]] mod 36. Apply it to 3-symbol blocks; preserve hyphens and spaces outside the cipher. A known-plaintext attacker can recover K by encrypting three independent basis blocks (100, 010, 001 in numeric form): their ciphertext vectors reveal the three columns of K. This demonstrates why linear Hill encryption is vulnerable when enough known plaintext is available.

TASK 6 – Evaluate Cryptanalytic Resistance (4 Marks)

Scheme	Suitable attack	Why it succeeds / fails	Resistance
Caesar	Brute force (all 26 shifts), frequency analysis	Tiny key space; letter-frequency pattern is unchanged.	Very low
Monoalphabetic	Frequency, digram and word-pattern analysis	One-to-one fixed substitution preserves frequency rankings and repeated patterns.	Low
Vigenère	Kasiski examination and index of coincidence	Repeated keyword causes periodic shifts; estimate key length, then solve each Caesar column.	Moderate only for long non-repeating keys; weak when repeated
Hill	Known-plaintext linear algebra	Ciphertext blocks give linear equations for K; enough independent plaintext vectors recover it.	Low against known plaintext
Columnar transposition	Anagram/digram and column-length analysis	Letter frequencies are exactly preserved; rearrangement patterns can be reconstructed.	Low

Index of coincidence (IC) is the probability that two randomly selected ciphertext letters are equal. English text is typically around 0.066; uniformly random letters are near $1/26 \approx 0.0385$. A monoalphabetic substitution preserves the frequency distribution and therefore preserves IC. A transposition also preserves the exact letter counts and IC. A Vigenère cipher with a sufficiently varied key tends to flatten frequencies, but repeated-key periodicity can be exposed by Kasiski or IC analysis.

Resistance ranking for these classical methods: Vigenère with a long non-repeating key (best of this group, but not a one-time pad unless the key is truly random, message-length, secret and never reused) > Hill for ciphertext-only attacks > columnar transposition > monoalphabetic substitution > Caesar. Against known plaintext, Hill falls sharply because of its linear structure. No classical method here is suitable for actual patient data.

TASK 7 – Assess Transposition and Product Ciphers (3 Marks)

Columnar transposition writes text into rows and reads columns in a key-defined order. It diffuses the positions of characters, but it does not change how many times each letter occurs. A substitution followed by transposition is a product cipher: substitution changes symbols and transposition changes positions. It is stronger than either component alone against simple visual inspection, but the classical components still have exploitable patterns and small or structured keys.

Evidence: transposition alone preserves the ciphertext histogram exactly. Monoalphabetic substitution also preserves the plaintext frequency counts, only relabelling the letters. Their combination can obscure both direct symbol identity and location, but frequency counts remain a relabelled version of plaintext counts. Known-plaintext and structural attacks remain possible.

Practicality: manual columnar transposition plus substitution requires staff to manage keys and avoid mistakes. For rural clinics with limited technical training, it is error-prone. A simple approved application is easier to use, but the underlying classical cipher is still weak.

TASK 8 – Evaluate Steganography (3 Marks)

LSB (least significant bit) image embedding stores report bits in the least significant bit of selected pixel channels. To embed, convert the report to bytes, add a length field, and replace the lowest bit of each selected channel with one message bit. Extraction reads those bits in the same order and reconstructs the report.

Criterion	Encryption	LSB steganography
Purpose	Makes content unreadable without a key.	Conceals the existence of a message.
Capacity	Depends on cipher and channel; ciphertext may be larger than plaintext due to padding/metadata.	Limited by available pixel channels and acceptable distortion.
Imperceptibility	Not applicable; ciphertext is visible as data.	Can be visually subtle at low embedding rates.
Robustness	Ciphertext survives copying exactly, but not arbitrary corruption.	Fragile; resizing, filtering or JPEG compression can destroy LSBs.
Detectability	Encrypted data is obvious as encrypted data.	Statistical tests such as chi-square can reveal unusual LSB patterns.

Medical-image risk: even a small pixel change may matter for diagnostic images, especially around subtle structures. Therefore, never alter the diagnostic original. If steganography is required, use a separate approved copy, keep the original unchanged, record provenance, and validate that the image remains clinically acceptable. Encrypt the report first and then embed the ciphertext (encrypt-then-embed). Hiding is not encryption: if the hidden report is extracted, it must still be unreadable without its key.

Text-based hiding can use a fixed pattern such as selecting the first letter after every agreed delimiter in a text circular. This has low capacity and is easy to detect if the pattern is known. It is not a robust secure channel.

TASK 9 – Implement and Test (4 Marks)

The attached Python program implements Caesar, Vigenère, a mod-36 3×3 Hill cipher, columnar transposition and basic LSB embedding/extraction. It includes round-trip tests. It preserves non-alphabet separators for the mod-36 Hill example. Run it using Python 3; no external package is required for the cipher tests. Pillow is required only for image steganography.

Reproducible procedure:

1. Save the accompanying file classical_crypto_assignment.py.
2. Run: python classical_crypto_assignment.py
3. Confirm that each test prints PASS and that decrypted plaintext equals the original.
4. To test image LSB, install Pillow using: python -m pip install pillow. Run: python classical_crypto_assignment.py embed input.png output.png "Short report" and then python classical_crypto_assignment.py extract output.png.
5. Repeat timing tests with the same generated plaintext at 100, 1,000, 10,000 and 100,000 letters. Record machine, Python version and run each case several times; report the median.

Edge cases to test: empty input, punctuation, digits in IDs, odd-length Hill input (pad with X and remove the pad after decryption), and messages whose length is not a multiple of the transposition column count. A production implementation would also need authenticated encryption, secure key storage and careful error handling.

TASK 10 – Record and Interpret Experimental Evidence (3 Marks)

The following are expected qualitative results from the mathematical properties of the implemented schemes. Exact runtime and attack-success measurements depend on the computer and must be filled using the supplied program; no machine-specific timings are invented here.

Measure	Caesar	Monoalphabetic	Vigenère	Hill	Columnar
Ciphertext letter histogram	Same counts, shifted labels	Same counts, relabelled	Flatter for varied key; periodic patterns remain	Block-dependent; may look flatter	Exactly same counts
Index of coincidence	Same as plaintext	Same as plaintext	Usually lower for varied key; rises when grouped by key period	Depends on block mixing; not a guarantee	Exactly same as plaintext
Encryption/decryption time	Very low	Very low	Low	Low to moderate	Low
Attack trend as text grows	Brute force unchanged; frequency	Frequency attack improves	Longer text helps Kasiski/IC	More independent blocks help	Longer text gives more rearrangement

	clearer		find period	known-plaintext recovery	clues

Graph instructions: create (1) plaintext and ciphertext letter-frequency bar charts for each cipher, (2) a bar chart of measured IC values, (3) a line graph of median encryption/decryption time against message length, and (4) a line graph of attack success rate against ciphertext length. Use identical plaintext inputs for all schemes. Interpret the graphs: Caesar, monoalphabetic and transposition retain frequency evidence; Vigenère masks single-letter frequencies but repeating-key evidence accumulates with longer messages; Hill's linear equations become easier to solve as independent known plaintext blocks increase.

TASK 11 – Recommend and Defend the Final Design (4 Marks)

Academic recommendation: combine role-based operational procedures with Bell–LaPadula confidentiality labels, use a different key for each clinic or session, rotate keys, avoid predictable fixed headers where possible, and use encrypt-then-embed only for a separate non-diagnostic image copy. For the classical-cipher comparison, Vigenère with a long non-repeating key plus transposition is the strongest layered classroom option among the selected methods, but it is not safe for real patient data.

Key distribution across 25 clinics is a major weakness. A single shared key means compromise at one clinic exposes messages from other clinics. Per-clinic keys reduce the blast radius but create storage, rotation, recovery and training costs. Keys must not be written on paper near terminals or shared among all staff.

Residual risks: classical ciphers do not provide modern authenticated encryption, reliable integrity, digital signatures, forward secrecy, secure key exchange or protection against a malicious authorized insider. BLP does not prevent a key-holding insider from copying plaintext or ciphertext, and it does not guarantee availability or integrity.

Final real-world judgement: classical ciphers are not acceptable for protecting actual patient records. A real deployment requires modern, vetted cryptography, authenticated key management, role-based access, multi-factor authentication, integrity protection, audit logging, backups, network security, incident response and legal/privacy compliance. The classical design is only an educational demonstration.

TASK 12 – Day-4 Design Challenge (8 Marks)

A. Impact analysis (2 Marks)

- The attacker is now a known insider at a satellite clinic, not merely an external listener. The attacker can collect many ciphertexts and may know the message formats.
- The assumption that all 25 clinics keep one shared key secret has failed. Key reuse and predictable headers such as PATIENT ID, DIAGNOSIS and PRESCRIPTION provide known-plaintext material.
- Large volumes make frequency analysis, Kasiski examination and known-plaintext attacks more reliable. Risk ranking increases for key compromise, prescription forgery and cross-clinic exposure.

B. Evaluation of original design (2 Marks)

A repeated-key Vigenère cipher is vulnerable when many messages use the same key. If a plaintext character is known, the key character is obtained by $K_i=(C_i-P_i)$ mod 26. For example, if a known header has P=A (0) and its ciphertext is D (3), the corresponding key character is D (3). Repeating known header positions across many messages reveals repeated key positions. For Hill, known independent plaintext blocks provide linear equations C=KP mod 36; enough independent blocks reveal the key matrix. Thus the original shared-key design is not defensible against this insider.

BLP still limits which records the insider may read through the application, provided access checks are correctly enforced. However, BLP is irrelevant once an insider legitimately obtains ciphertext and the shared decryption key: the model cannot make the key secret from its holder or prevent copying outside the controlled system.

C. Adaptation or redesign (2 Marks)

6. Replace one network-wide key with unique per-clinic keys and short-lived per-session keys.
7. Rotate keys on a defined schedule and immediately revoke a clinic key after suspected compromise.
8. Use randomised padding or variable message framing to reduce predictable headers; do not rely on header hiding as the main defence.
9. Separate encryption keys by purpose (clinical records, prescriptions and billing).
10. If a hidden report is required, encrypt it first and embed only in a separate approved image copy. Never modify the diagnostic source image.
11. Keep BLP labels, add break-glass auditing and use trusted declassification for billing.

These changes reduce the number of ciphertexts available under one key and limit the impact of a single clinic compromise. They do not make classical algorithms modern-secure; changing keys reduces exposure but does not repair weak cipher designs.

D. Evidence-based defence (2 Marks)

The mod-26 Caesar key space is only 26; monoalphabetic substitution has 26! keys but leaks frequency; repeating Vigenère has 26^m keys but repeated keys and known headers expose periodic structure; Hill is linear and recoverable from sufficient known plaintext; transposition preserves the histogram. Per-clinic/session keying reduces reuse and cross-clinic damage, but increases key-management workload. Staff usability requires centrally managed software, clear login steps, training and emergency access procedures.

Final judgement: retain BLP as a confidentiality policy, modify operational controls and key separation, but redesign the cryptographic protection for any real deployment. No classical cipher can guarantee modern integrity, authenticated origin, secure key exchange or protection from an authorized insider who can access plaintext.

Conclusion

The healthcare network needs confidentiality, integrity, authentication, availability and accountability. Classical ciphers illustrate important mathematical ideas but are vulnerable to brute force, frequency analysis, repeated-key attacks and known-plaintext attacks. Bell–LaPadula helps control information flow but does not protect integrity or solve emergency access by itself. The safest academic

recommendation is layered access control, per-clinic/session key separation, auditing and careful handling of hidden reports. For real patient data, classical ciphers must be replaced with modern authenticated cryptography and professional key management.

Appendix A – Python implementation

The separate source file accompanies this report. The following commands demonstrate the intended use:

```
python classical_crypto_assignment.py
python classical_crypto_assignment.py embed scan.png scan_with_report.png "LAB REPORT: SAMPLE"
python classical_crypto_assignment.py extract scan_with_report.png
```

Appendix B – References

- CO1 Assignment 1 PDF supplied with the question paper (scenario, tasks and marking scheme).
- Bell, D. E. and LaPadula, L. J., Secure Computer Systems: Mathematical Foundations and Model, MITRE technical report (1973).
- Stallings, W., Cryptography and Network Security: Principles and Practice, Pearson.
- Shannon, C. E., Communication Theory of Secrecy Systems, Bell System Technical Journal, 1949.

Supplementary Evidence for the Extended Design Requirements

1. Twelve Bell–LaPadula access decisions

The following additional decisions expand the access-control exercise to more than twelve requests. A subject may read objects at its own level or lower, but not higher. A subject may write at its own level or higher, but not lower.

Access request	Decision	BLP reasoning
Medical Director TS reads Restricted invoice	Allow	Read down
Medical Director TS writes Secret specialist note	Allow	Write up
Specialist S reads Top Secret strategic file	Deny	Read up
Specialist S writes Confidential nursing note	Deny	Write down
Specialist S writes Top Secret referral record	Allow	Write up
Nurse C reads Restricted billing code	Allow	Read down
Nurse C reads Secret allergy record	Deny	Read up
Nurse C writes Secret clinical escalation	Allow	Write up
Lab staff C writes Public patient leaflet	Deny	Write down
Billing staff R reads Public patient instructions	Allow	Read down
Billing staff R reads Confidential lab result	Deny	Read up
Patient P reads Public discharge summary	Allow	Same level
Patient P writes Restricted billing record	Allow	Write up
Patient P reads Restricted invoice detail	Deny	Read up

2. Lab-table transposition and checksum test

Example: a 20-row lab table contains columns PatientCode, TestCode, Value and Unit. A columnar transposition may rearrange the serialized table text for confidentiality, but it does not detect modification. For a simple teaching checksum, map A=1…Z=26 and digits to their numeric values, then calculate S = (sum of all serialized character values) mod 97. Store S separately with the table.

Test: calculate S for the original 20-row table, then change one result cell (for example, 5.6 to 8.6) and recalculate. The checksum will usually change and the alteration is detected. If two cells or rows are

swapped, a simple sum checksum may remain identical because the same characters are present. An attacker who can modify the table and recompute the unkeyed checksum can also defeat it. Therefore, this checksum is only an error-detection demonstration, not a secure integrity mechanism.

3. Classical medical-image encryption experiment

For an 8-bit grayscale image I(x,y), an intensity shift is $E(x,y) = (I(x,y)+k) \bmod 256$; decryption is $D(x,y) = (E(x,y)-k) \bmod 256$. It changes pixel values but applies the same shift everywhere, so the image's spatial structure and edges remain visible. Its histogram is circularly shifted; entropy is unchanged because the mapping is a one-to-one permutation of the 256 intensity values.

Pixel permutation uses a secret permutation π of pixel positions: $E[\pi(i)] = I[i]$. Decryption applies π^{-1}. It preserves the exact histogram and entropy, but spatial adjacency is destroyed, so the image's visual structure is less directly visible. A fixed permutation is still vulnerable to known/chosen-plaintext analysis and does not provide authentication.

For both methods, test on a 256×256 or larger grayscale image, decrypt, and compare the decrypted image byte-for-byte with the original. Report histogram and Shannon entropy $H = -\Sigma\ p(i)\ \log_2 p(i)$. Do not use patient diagnostic images without authorization; use a public sample image.

4. Steganography measurements

For LSB embedding, capacity with one bit per grayscale pixel is width×height bits (for RGB, up to 3×width×height bits if all channels are used). A length field and metadata reduce usable capacity. PSNR is $10 \log_{10}(MAX^2/MSE)$, where MSE is the mean squared pixel error and MAX=255 for 8-bit images. For one LSB change per modified channel, MSE is small and PSNR is often high, but the numerical value depends on the image and payload.

A chi-square test compares observed counts of paired intensity values (2k and 2k+1) with expected equal counts; strong LSB replacement can make pairs unusually balanced and therefore detectable. JPEG compression is lossy and usually changes pixel LSBs, so basic LSB payloads may not survive. Test by embedding in PNG, extracting before compression, saving/re-encoding as JPEG, then attempting extraction; report whether the message is recovered. Do not claim robustness if extraction fails.

Text-circular hiding demonstration: write a short report using the first letter of each sentence in a harmless circular. The recipient extracts the acrostic. It has very low capacity, is easy to notice if sentence initials are inspected, and ordinary editing can destroy it. It is not a replacement for encryption.

5. How to complete the experimental graphs accurately

Run the supplied program for its implemented ciphers and record the actual median timings from your own computer. For a fair comparison, use the same generated plaintext lengths (100, 1,000, 10,000 and 100,000 characters), repeat each test at least five times, and report the median. The current source code times the Caesar implementation as a reproducible baseline; extend the timing loop to call Vigenère, Hill and columnar functions before claiming a measured cross-scheme comparison.

For frequency histograms and IC, use the same plaintext for every algorithm and exclude punctuation consistently. For attack success, define success clearly (for example, correct key or recovered plaintext), test each attack on multiple ciphertext lengths, and report successes divided by trials. Do not invent numerical success rates: the result depends on the chosen messages, keys and attack implementation.

DETAILED NOTES 01: Expanded Introduction: Why Patient-Record Security Matters

Healthcare information is unusually sensitive because it describes a person's health, treatment, identity and financial relationship with a provider. A record may include a diagnosis, medication, allergy, laboratory value, scan report, address and billing details. If the information is disclosed, a patient may experience embarrassment, discrimination or financial harm. If it is altered, the result may be more serious: a clinician may prescribe the wrong medicine, overlook an allergy or make a decision using an incorrect laboratory value.

The network in this assignment has one central hospital and 25 satellite clinics. The public network is a shared communication environment, so messages may be observed while travelling between locations. The clinics may also have different levels of staff experience and computing resources. A security design must therefore be understandable, repeatable and manageable. A mathematically complicated method that staff cannot operate correctly may create new risks.

The assignment deliberately restricts the design to classical ciphers and access-control reasoning covered in CO1. This is useful for learning the foundations of encryption, modular arithmetic, substitution, transposition, cryptanalysis and information-flow policy. It does not mean that these methods are suitable for real healthcare deployment. The final recommendation must clearly separate classroom demonstrations from real security requirements.

The most important design principle is that confidentiality and correctness are different requirements. A cipher may conceal a message while it travels, but it does not automatically prove that the message was created by the correct doctor. It also does not automatically detect a modified dosage. Similarly, access control can restrict who reads a record, but it cannot guarantee that the record itself is accurate.

The system must protect three broad kinds of information: clinical records, operational records and patient-facing information. Clinical records include diagnoses, prescriptions, laboratory results and specialist opinions. Operational records include appointments, staff activity and billing. Patient-facing information includes approved summaries and instructions. These categories should not be mixed without a deliberate decision about who needs to see each item.

For the rest of the report, each method is judged by its purpose, mathematical structure, likely attacks, usability and limitations. Worked examples use fictional text only. The same examples can be reproduced using the Python program supplied with this assignment.

Assignment evidence / study point: Relate this discussion to the healthcare network, identify the assumption being made, and state the limitation of the classical method before drawing a conclusion.

DETAILED NOTES 02: Healthcare Network Architecture and Data Flow

The central hospital acts as the coordination point for the 25 clinics. A clinic collects patient information and sends it to the central hospital. The central hospital may store the record, request a specialist opinion, return treatment advice or send a prescription-related response. Billing and front-desk staff may need administrative details, but they do not need unrestricted access to clinical diagnoses.

A simple data flow can be described in six stages. First, an authorized employee logs in at the clinic. Second, the employee selects the patient and the record type. Third, the application checks the employee's clearance and the requested operation. Fourth, the record is prepared for transmission. Fifth, the receiving hospital checks the sender and processes the message. Sixth, the event is recorded for later review. In the classroom model, classical encryption illustrates the transformation of plaintext into ciphertext; BLP illustrates the access decision.

The public network creates opportunities for eavesdropping, traffic analysis, replay and message modification. A clinic workstation creates a different risk: an insider may copy files or share a key. The central hospital creates a concentration risk because a large number of records are stored in one location. A useful design therefore considers endpoints, users, keys, messages and stored records rather than focusing only on the cipher.

The architecture should separate user identity from classification. A person may be a nurse, but the system should still verify the person's account and current role. A role describes job responsibility; a security level describes the maximum information classification that may be accessed under the policy. Both matter.

A basic text-based workflow could use the following checks: Is the account active? Is the employee assigned to this clinic? Does the employee's clearance permit the requested read? Does the requested write obey the *-property? Is the record related to the employee's assigned task? Is emergency access being requested? These questions are more useful than assuming that everyone inside a hospital network is trusted.

The network must also plan for unavailable connections. Rural clinics may temporarily lose connectivity. A safe design should define how urgent care proceeds, how local records are protected, and how updates are synchronized later. Classical ciphers do not solve availability. This is why availability must be treated as a separate security goal and not as a side effect of encryption.

Assignment evidence / study point: Relate this discussion to the healthcare network, identify the assumption being made, and state the limitation of the classical method before drawing a conclusion.

DETAILED NOTES 03: Detailed Threat, Vulnerability, Attack and Risk Analysis

The terms threat, vulnerability, attack and risk are related but should not be used as synonyms. A threat is a potential source of harm, such as a dishonest employee or an attacker monitoring a public network. A vulnerability is a weakness, such as a shared key, weak password, unpatched workstation or predictable message format. An attack is an action that exploits a weakness, such as copying ciphertext, impersonating a doctor or changing a dosage. Risk combines the chance of the event with the seriousness of its consequences.

For example, an insider who has access to a clinic's shared Vigenère key is a threat actor. Reusing that key for hundreds of messages is a vulnerability. Collecting ciphertexts and using known headers to infer the key is an attack. The risk is that diagnoses or prescriptions may be exposed across many records. This example shows why a single weakness can affect a large number of patients.

A useful risk assessment considers both privacy and patient safety. A disclosure may be harmful, but a modified prescription can lead directly to an incorrect treatment. The impact rating should therefore reflect clinical consequences, not only the amount of data exposed.

The risk score in this report is calculated as likelihood multiplied by impact. A score from 1 to 4 is low, 5 to 9 is moderate, 10 to 16 is high, and 17 to 25 is critical. These bands are a simple classroom scale, not a universal healthcare risk standard. Likelihood estimates how easily the attack may occur in the described environment. Impact estimates the harm if it succeeds.

The ranking should be revisited when the threat model changes. Before the Day-4 challenge, an external listener may be the main concern. After the challenge, an insider has already copied hundreds of ciphertexts under the same key. The likelihood of key-related cryptanalysis becomes higher because the attacker has more material and predictable message formats. The potential impact also increases if the key is shared across multiple clinics.

Risk treatment includes prevention, detection, response and recovery. Prevention may include limiting key reuse and enforcing access rules. Detection may include audit logs and review of unusual downloads. Response may include revoking a compromised key. Recovery may include restoring correct records and contacting affected clinical teams. A risk table is useful only when it leads to specific actions.

Assignment evidence / study point: Relate this discussion to the healthcare network, identify the assumption being made, and state the limitation of the classical method before drawing a conclusion.

DETAILED NOTES 04: Passive Attacks in the Healthcare Scenario

A passive attack attempts to learn information without changing the data or disrupting the service. Eavesdropping is the simplest example. An attacker copies network traffic between a satellite clinic and the central hospital. If the cipher is weak or the key is exposed, the attacker may recover patient names, diagnoses or prescriptions. Even when the message content remains encrypted, the attacker may learn which clinic communicates with the specialist centre and how frequently.

Traffic analysis studies communication patterns. The attacker may observe the time, size, frequency and endpoints of messages. A sudden increase in messages from one clinic may suggest an emergency, outbreak investigation or specialist referral. A large scan upload may reveal that a complex diagnostic case is being handled. Traffic analysis can therefore disclose sensitive context without recovering the plaintext.

Ciphertext collection is also important. The Day-4 insider copies hundreds of encrypted messages. The attacker may not immediately decrypt them, but a large collection provides statistical evidence. Repeated headers and reused keys create patterns that can be compared across messages. Classical encryption often leaks structure, so the amount of available ciphertext can make cryptanalysis more effective.

Passive attacks are difficult to notice because the attacker may not alter any record. A network may continue operating normally while information is copied. This makes prevention and monitoring important. Staff should not assume that the absence of visible errors proves that data has remained private.

The classroom ciphers have different leakage patterns. Caesar and monoalphabetic substitution preserve letter-frequency relationships. Columnar transposition preserves the exact letter counts. Vigenère can flatten single-letter frequencies when the key is sufficiently varied, but a repeating keyword introduces periodic structure. Hill encryption mixes letters in blocks, but its linear equations can be exposed with enough known plaintext.

The best response is not to claim that one classical cipher removes all passive-attack risk. Instead, the report compares what each method leaks and explains why repeated-key use is dangerous. In a real system, modern authenticated encryption and managed keys are required. Classical methods remain valuable here as a way to understand why frequency patterns and key reuse matter.

Assignment evidence / study point: Relate this discussion to the healthcare network, identify the assumption being made, and state the limitation of the classical method before drawing a conclusion.

DETAILED NOTES 05: Active Attacks in the Healthcare Scenario

An active attack changes data, impersonates a user, replays an old message or disrupts service. Prescription modification is a high-impact example. An attacker changes the medicine name, strength, frequency or duration. The receiving clinic may decrypt the altered message successfully because decryption does not prove that the ciphertext is authentic. A valid-looking plaintext can still be false.

Masquerade occurs when an attacker pretends to be an authorized doctor or clinic. If the network relies only on a shared secret key, anyone who obtains that key may create ciphertexts that look legitimate. Replay occurs when an attacker resends an old valid prescription or specialist order. Even if the original message was genuine, it may no longer be appropriate for the patient or current treatment.

Denial of service makes the records system unavailable. The attacker may flood a server, disrupt connectivity or damage a workstation. In rural settings, even a short outage may delay access to a specialist opinion. Classical ciphers do not prevent service disruption, so availability needs separate operational controls.

Active attacks require more than confidentiality controls. The receiver should be able to determine who sent the message, whether it is fresh, whether it has been changed and whether it is intended for the correct patient. In a classroom assignment, these requirements can be discussed as gaps in classical cipher design. A classical cipher alone cannot meet them.

A replay-resistant message would need contextual information such as a unique order number, timestamp or sequence number, and the receiver would need to check it. However, an unprotected timestamp can itself be modified. In a real system, freshness information must be authenticated together with the message.

For prescription integrity, a checksum can detect accidental changes, but an attacker may recompute an unkeyed checksum. A cryptographic message authentication mechanism or digital signature is required in real deployment. The assignment's classical checksum exercise is intended to demonstrate this limitation, not to recommend a checksum for clinical use.

Assignment evidence / study point: Relate this discussion to the healthcare network, identify the assumption being made, and state the limitation of the classical method before drawing a conclusion.

DETAILED NOTES 06: Risk Ranking: Worked Assessment and Treatment

The risk table uses a five-point scale. Likelihood 1 means rare, 3 means possible and 5 means very likely in the scenario. Impact 1 means minor inconvenience, 3 means meaningful privacy or operational harm and 5 means severe patient-safety or network-wide consequences. The score is the product of the two ratings.

Prescription modification receives a critical score because the consequence can be immediate physical harm. Masquerade as a doctor is also critical because false orders may be trusted. Replay is critical where an old instruction can be administered again. Eavesdropping is likely on a public network, but its direct impact depends on the sensitivity of the data and whether the cipher is broken. Traffic analysis is a privacy risk even when the message remains unreadable. Denial of service can be serious if it occurs during urgent care.

The order of treatment should be based on expected harm. First, protect the correctness and origin of prescriptions. Second, control key compromise and impersonation. Third, detect replay and unusual message activity. Fourth, reduce exposure from passive monitoring. Fifth, strengthen availability and recovery so clinics can continue essential care.

This ordering does not mean confidentiality is unimportant. It means patient safety must be considered alongside privacy. A design that hides every diagnosis but allows a forged dosage is not a successful healthcare design. A design that prevents unauthorized reads but blocks all emergency access is also incomplete.

Risk treatment should be assigned to owners. Clinic administrators manage user access and staff training. Central IT manages key rotation and system monitoring. Clinical governance defines emergency access and reviews unusual prescription events. The central hospital coordinates incident response across all clinics. This division makes the recommendations actionable.

Assignment evidence / study point: Relate this discussion to the healthcare network, identify the assumption being made, and state the limitation of the classical method before drawing a conclusion.

DETAILED NOTES 07: Confidentiality, Integrity and Availability in Detail

Confidentiality means that information is available only to authorized people. In the scenario, billing staff should not automatically read a sensitive diagnosis merely because they can view an invoice. Classical substitution and transposition ciphers attempt to conceal message content, but they are weak because their patterns can be attacked and keys may be reused.

Integrity means that information remains accurate and complete. A prescription should not change from 250 mg to 500 mg without an authorized clinical action. Classical encryption alone does not provide integrity. A receiver can decrypt altered ciphertext and obtain altered plaintext. A simple checksum can detect some accidental changes, but it can be recalculated by an attacker.

Availability means that authorized users can obtain records when needed. A clinic may need a specialist's opinion during an emergency. Encryption does not guarantee that the network, server or workstation is functioning. Backups, redundant connections, recovery plans and local emergency procedures are separate requirements.

Authentication verifies an identity or the source of a message. A shared key does not provide strong individual attribution because every holder of the key may produce a message. Non-repudiation means that a sender cannot credibly deny an action later; classical shared-key ciphers do not provide this because the key is shared.

The goals interact. Strong confidentiality with poor availability may delay treatment. Strong access restrictions with poor emergency procedures may prevent a nurse from seeing an allergy. Integrity without authentication may protect a record from accidental changes but not prove who created it. Therefore, the system needs a balanced set of controls.

The assignment asks students to map goals to record upload, prescription transmission and specialist consultation. The mapping should be justified by the type of harm each function could create. Prescription transmission places integrity and authentication at the top because dosage and sender identity are safety-critical.

Assignment evidence / study point: Relate this discussion to the healthcare network, identify the assumption being made, and state the limitation of the classical method before drawing a conclusion.

DETAILED NOTES 08: Bell–LaPadula: Subjects, Objects and Labels

A subject is an active entity that requests access, such as a doctor, nurse or billing employee. An object is a resource being accessed, such as a patient record, diagnosis file, laboratory result or invoice. A security label assigns a classification to the subject or object. The model compares these labels before allowing an operation.

The assignment defines five levels: Top Secret, Secret, Confidential, Restricted and Public. The Medical Director is Top Secret. Specialist Doctors are Secret. Nurses and Lab Staff are Confidential. Billing and Front-Desk Staff are Restricted. Patients are Public. These labels provide a simplified hierarchy for the classroom example.

A subject's clearance is not the same as a clinical need-to-know. A doctor may have Secret clearance but should still access only records relevant to assigned patients. BLP models information-flow restrictions; a separate role or purpose check is needed to limit access to the correct patient and task.

The object labels should reflect the sensitivity of their contents. A diagnosis may be Secret or Top Secret depending on the assignment's classification policy. Routine lab results may be Confidential. Billing details may be Restricted. A patient-facing discharge summary may be Public after it has been reviewed and approved for release.

Classification must be consistent. If the same diagnosis is stored in a high-level clinical record and copied into a lower-level billing note, the lower-level copy may disclose information that the original policy intended to protect. Declassification must therefore be deliberate and limited to the minimum necessary information.

The hierarchy in this assignment is an educational simplification. Actual healthcare access is often more complex and may involve patient consent, care-team membership, jurisdiction, emergency status and purpose of use. These factors can be discussed as additional controls beyond the basic BLP model.

Assignment evidence / study point: Relate this discussion to the healthcare network, identify the assumption being made, and state the limitation of the classical method before drawing a conclusion.

DETAILED NOTES 09: Bell–LaPadula: Simple Security Property

The simple security property is often summarized as “no read up.” A subject cannot read an object at a higher classification than the subject's clearance. A Secret-level specialist can read Secret, Confidential, Restricted and Public information, but cannot read Top Secret information. A Confidential-level nurse can read Confidential, Restricted and Public information, but cannot read Secret or Top Secret information.

Worked example: a nurse at Confidential level requests a Secret-level allergy record. The requested object is above the nurse's clearance. The system denies the read. The reason is not that the nurse is untrustworthy; it is that the model prevents information from flowing from a higher classification to a lower-cleared subject.

Another example: a billing clerk at Restricted level requests a Confidential laboratory result. The result is higher than Restricted, so the request is denied. The clerk may receive only the billing information that has been appropriately released.

The no-read-up rule helps limit disclosure, but it can create difficulty in healthcare. A nurse may genuinely need a high-level allergy record to provide safe treatment. The model does not understand clinical urgency by itself. It applies the classification rule and denies the request unless a separate emergency mechanism authorizes a controlled exception.

This illustrates why access control must be designed around both security and patient care. A rigid rule may preserve confidentiality but cause delay. A broad exception may improve speed but become an opportunity for misuse. Emergency access should be narrow, temporary, logged and reviewed.

For each access decision, the report should state the subject level, object level, operation and rule applied. This makes the result auditable and avoids simply writing “allow” or “deny” without explanation.

Assignment evidence / study point: Relate this discussion to the healthcare network, identify the assumption being made, and state the limitation of the classical method before drawing a conclusion.

DETAILED NOTES 10: Bell–LaPadula: Star Property and Information Flow

The *-property is commonly summarized as “no write down.” A subject cannot write information to an object at a lower classification. A Secret-level specialist may write to a Secret or Top Secret object, but not directly to a Confidential, Restricted or Public object. This rule prevents high-classification information from being copied into a lower-classification location.

Worked example: a Secret-level doctor tries to place a detailed diagnosis in a Restricted billing file. The request is denied because the destination is lower than the subject's level. If the doctor writes the diagnosis directly into the invoice notes, the information could become visible to staff who should not see it.

A controlled declassification process may create a limited billing statement containing only a procedure code, approved amount and payment status. The process should not copy the full diagnosis. The released information is reviewed and intentionally downgraded.

The *-property may initially seem inconvenient because a doctor cannot write a lower-level note. However, the rule is designed to prevent information from flowing downward without review. A trusted subject or approved process may perform a controlled release, but the exception must be narrow and accountable.

A subject writing at a higher level is generally permitted under the simple model. This can create a different operational problem: a nurse may need to enter a clinical observation into a higher-level record. The policy must specify which clinical application and trusted workflow can perform the operation.

The model's strength is its simple information-flow reasoning. Its weakness is that it does not judge whether the content is medically correct. A doctor with appropriate clearance may still enter an incorrect dosage. BLP must therefore be combined with integrity and clinical validation controls.

Assignment evidence / study point: Relate this discussion to the healthcare network, identify the assumption being made, and state the limitation of the classical method before drawing a conclusion.

DETAILED NOTES 11: Bell–LaPadula: Emergency Access and Break-Glass

Emergency access is needed when ordinary permissions would delay urgent treatment. Consider a nurse who needs to view a Secret-level allergy record while treating an unconscious patient. Under normal BLP rules, the Confidential-level nurse cannot read up. A break-glass process can provide temporary access when the emergency is genuine.

A responsible break-glass process should require the user to identify the patient, select an emergency reason and confirm that ordinary access is insufficient. The system should grant only the minimum necessary access for a limited period. It should record the employee, patient, data accessed, time, reason and actions performed. The event should be reviewed by an authorized supervisor after the emergency.

Break-glass must not become a routine shortcut. Repeated use by the same employee, access to unrelated patients or access outside working hours may require investigation.

Emergency access also needs a safe fallback when the network is unavailable. A clinic may need a locally available emergency summary, but that summary must be protected and kept current. A printed emergency record may be useful in some circumstances, but it introduces physical security and update risks.

The policy should define who can authorize emergency access, how long it lasts, which records are included and how the event is reviewed. A nurse should not receive permanent Secret clearance merely because emergencies can occur. The exception should be temporary and traceable.

The BLP model alone does not define a complete emergency workflow. It provides the baseline rule that the workflow must deliberately override. This is an important limitation to discuss in the assignment's critical evaluation.

Assignment evidence / study point: Relate this discussion to the healthcare network, identify the assumption being made, and state the limitation of the classical method before drawing a conclusion.

DETAILED NOTES 12: Tranquility, Trusted Subjects and BLP Limitations

The tranquility principle concerns changes to security labels while a system is operating. Strict tranquility prevents labels from changing during operation. This makes the policy easier to reason about because a subject or object cannot silently move to a different level. A relaxed approach permits controlled changes under defined conditions, but it requires careful authorization.

Trusted subjects are specially authorized entities that may perform actions that ordinary subjects cannot, such as declassification. Because these entities can bypass ordinary information-flow restrictions, they must be limited, monitored and audited. A trusted process should release only the information required for the lower-level purpose.

BLP is a confidentiality model. It does not guarantee integrity, correctness, availability or identity authentication. It may also be affected by covert channels, where information is communicated through indirect system behavior such as timing or resource use.

Healthcare needs more than a confidentiality lattice. A patient record can be confidential but wrong. A prescription can be visible only to the doctor and nurse but still contain an incorrect dosage. A record system can enforce read restrictions but become unavailable during an outage. These problems are outside BLP's main purpose.

The model can also be too rigid for multidisciplinary care. A specialist, nurse, laboratory employee and billing clerk may need different parts of one patient's record. A single label may not represent all the contextual permissions required. Compartments, roles, patient relationship and purpose-of-use checks can make the policy more realistic.

The conclusion is that BLP is useful for demonstrating mandatory confidentiality control, but it is not a complete healthcare security architecture. The final design should combine it with other safeguards.

Assignment evidence / study point: Relate this discussion to the healthcare network, identify the assumption being made, and state the limitation of the classical method before drawing a conclusion.

DETAILED NOTES 13: Classical Cryptography: Core Terminology

Cryptography is the study of methods for protecting information. Plaintext is the readable message before encryption. Ciphertext is the transformed message. Encryption converts plaintext into ciphertext using an algorithm and key. Decryption converts ciphertext back into plaintext using the corresponding decryption process and key. Cryptanalysis studies methods for recovering plaintext or keys without being given the secret key.

A symmetric cipher uses the same secret key, or closely related secret information, for encryption and decryption. The sender and receiver must both protect the key. If the key is copied by an insider, confidentiality may be lost. The healthcare challenge demonstrates this problem through an employee who has collected many ciphertexts under a reused key.

Classical ciphers are commonly divided into substitution and transposition methods. Substitution replaces symbols with other symbols. Transposition rearranges the positions of symbols. Polyalphabetic substitution uses more than one substitution alphabet, often controlled by a repeating keyword.

The security of a cipher depends on more than the secrecy of its algorithm. The key should be secret, the key space should be large enough for the threat, and the algorithm should resist known attacks. Classical methods often fail because they preserve language statistics or use linear transformations that can be solved from known examples.

The terms confusion and diffusion are often used to explain cipher design. Confusion makes the relationship between key and ciphertext complicated. Diffusion spreads the influence of plaintext structure across ciphertext positions. Classical substitution and transposition illustrate early forms of these ideas, but their designs are not strong enough for modern protection.

In this assignment, formulas use modular arithmetic. Modulo 26 maps letters A–Z to numbers 0–25. Modulo 36 extends the alphabet to letters and digits. The modulo operation wraps values around after reaching the size of the alphabet.

Assignment evidence / study point: Relate this discussion to the healthcare network, identify the assumption being made, and state the limitation of the classical method before drawing a conclusion.

DETAILED NOTES 14: Caesar Cipher: Full Working and Example

The Caesar cipher is a monoalphabetic substitution cipher. Each plaintext letter is shifted by a fixed number of positions. Let A=0, B=1, …, Z=25 and let k be the shift key. Encryption is $C=(P+k) \bmod 26$. Decryption is $P=(C-k) \bmod 26$. A shift of 3 maps A to D, B to E and so on; letters near the end wrap around to the beginning.

Example: plaintext PATIENTID and key k=3. P maps to S, A to D, T to W, I to L, E to H, N to Q, T to W, I to L and D to G. The ciphertext is SDWLHQWLG. Decryption subtracts 3 from each ciphertext letter and recovers PATIENTID.

The Caesar cipher is easy to implement and useful for demonstrating encryption and decryption. It is not appropriate for real records because the key space is tiny and an attacker can test every possible shift.

There are 26 possible shift values if the identity shift is included, or 25 meaningful non-identity shifts. Brute force is therefore immediate. An attacker can decrypt the same ciphertext using every shift and recognize likely English words such as PATIENT, DOCTOR or DIAGNOSIS.

Frequency analysis also works because every plaintext letter is always changed to the same ciphertext letter. The most frequent ciphertext symbol may correspond to E, T or A. Common short words and repeated patterns provide additional clues.

The Caesar cipher does not hide message length. If punctuation and spaces are preserved, the word pattern is also visible. If spaces are removed, the letter-frequency weakness remains. Its educational value is that it clearly demonstrates a keyed substitution function and the difference between encryption and security.

Assignment evidence / study point: Relate this discussion to the healthcare network, identify the assumption being made, and state the limitation of the classical method before drawing a conclusion.

DETAILED NOTES 15: Monoalphabetic Substitution: Key Space and Weakness

A monoalphabetic substitution cipher uses a fixed permutation of the alphabet. The sender chooses a scrambled alphabet and replaces every plaintext letter with its corresponding ciphertext letter. If A maps to Q, every A in the plaintext becomes Q. The same mapping is used throughout the message.

There are 26 choices for the ciphertext symbol assigned to the first plaintext letter, 25 choices for the next, 24 for the next, and so on. The total key space is 26!, approximately 4.03×10^{26} possible permutations. This is much larger than the Caesar key space.

However, the cipher preserves the pattern of letter frequencies. The most common plaintext letter remains the most common ciphertext letter, although its identity changes. Repeated words also retain repeated patterns. For example, a repeated three-letter word produces the same three-letter ciphertext pattern every time.

An attacker does not need to test all 26! keys. Frequency analysis can estimate likely mappings. Digram and trigram analysis examines pairs and groups of three letters. Common English patterns such as TH, HE, IN, ER and THE can help identify substitutions. Word lengths and repeated-letter patterns also provide evidence.

This is an important example of why a large key space does not guarantee security. The key space measures the number of possible keys, but the attacker's job may be much easier if the ciphertext leaks statistical information.

In the healthcare scenario, predictable terms such as PATIENT, DIAGNOSIS and PRESCRIPTION make the attack easier. If the attacker knows the likely plaintext format, the substitution mapping can be inferred more quickly.

Assignment evidence / study point: Relate this discussion to the healthcare network, identify the assumption being made, and state the limitation of the classical method before drawing a conclusion.

DETAILED NOTES 16: Vigenère Cipher: Encryption, Decryption and Key Repetition

The Vigenère cipher is a polyalphabetic substitution cipher. Instead of using one fixed shift, it uses a repeating keyword to select a different shift for each letter. Convert A–Z to 0–25. For plaintext letter P_i and key letter K_i, encryption is $C_i=(P_i+K_i) \bmod 26$. Decryption is $P_i=(C_i-K_i) \bmod 26$.

Example: plaintext PATIENTDIAGNOSIS and keyword CARE. Repeat the keyword to the length of the plaintext: CARECARECARECARE. The ciphertext is RAKMGNKHKAXRQSZW. Decrypting with the same repeating key returns the original plaintext.

Vigenère can hide simple single-letter frequency patterns better than Caesar because the same plaintext letter can encrypt to different ciphertext letters depending on the key position. However, the keyword repeats, creating periodic behavior.

The key space for a keyword of fixed length m is 26^m. This grows quickly as m increases. But a long key space alone is not enough. If the key is repeated and the attacker has many ciphertexts, the attacker can estimate the period and then analyze each position group as a Caesar cipher.

Kasiski examination looks for repeated ciphertext sequences and measures the distances between them. Common factors of these distances may suggest the key length. The index of coincidence compares letter-frequency concentration. If ciphertext is grouped by the likely key period, each group may show language-like frequency patterns.

The Day-4 insider has many messages and predictable headers. This makes Vigenère less resistant than it may appear in a short demonstration. Reusing the same keyword across many clinic messages is a serious weakness.

Assignment evidence / study point: Relate this discussion to the healthcare network, identify the assumption being made, and state the limitation of the classical method before drawing a conclusion.

DETAILED NOTES 17: Hill Cipher: Matrix Encryption and Decryption

The Hill cipher is a block substitution cipher based on matrix multiplication. A block of n letters is converted into a column vector P. An n×n key matrix K is multiplied by P modulo 26. Encryption is $C=KP \bmod 26$. Decryption requires the inverse matrix K^{-1} so that $P=K^{-1}C \bmod 26$.

For a 2×2 example, use K=[[3,3],[2,5]]. Its determinant is $3\times5-3\times2=9$. Since $\gcd(9,26)=1$, the matrix is invertible modulo 26. The inverse of 9 modulo 26 is 3 because $9\times3=27$, which is congruent to 1 modulo 26.

The adjugate matrix is [[5,−3],[−2,3]]. Multiplying by 3 and reducing modulo 26 gives K^{-1}=[[15,17],[20,9]].

Encrypt the block HI. H=7 and I=8, so $P=[7,8]^T$. Multiply K by P: the first value is $3\times7+3\times8=45$, and the second is $2\times7+5\times8=54$. Modulo 26, these become 19 and 2. The values 19 and 2 correspond to T and C, so HI encrypts to TC.

To decrypt, multiply $[19,2]^T$ by K^{-1}. The first value is $15\times19+17\times2=319$, which is 7 modulo 26. The second is $20\times19+9\times2=398$, which is 8 modulo 26. The result is HI.

The Hill cipher mixes letters within a block, so it can hide repeated single-letter patterns better than a monoalphabetic substitution. However, it is linear. Enough independent known plaintext and corresponding ciphertext blocks can reveal the key matrix.

Assignment evidence / study point: Relate this discussion to the healthcare network, identify the assumption being made, and state the limitation of the classical method before drawing a conclusion.

DETAILED NOTES 18: Extended Hill Cipher: Modulo 36 for IDs and Dosages

The assignment requires an alphabet that includes letters and digits. Define the alphabet as ABCDEFGHIJKLMNOPQRSTUVWXYZ0123456789. This contains 36 symbols, mapped to values 0 through 35. Use blocks of three symbols and perform arithmetic modulo 36. Hyphens and spaces can be preserved outside the encrypted character stream so that identifiers remain readable in their overall format.

An example invertible 3×3 matrix is K=[[1,2,0],[0,1,0],[0,0,1]]. Its determinant is 1. Since gcd(1,36)=1, it has an inverse modulo 36. The inverse is [[1,34,0],[0,1,0],[0,0,1]], because 34 is congruent to −2 modulo 36.

For a block vector $[p_1,p_2,p_3]^T$, encryption gives $[p_1+2p_2,p_2,p_3]^T$ modulo 36. Decryption subtracts $2p_2$ from the first component. This is a deliberately simple key for demonstrating invertibility.

For the string PT20481, the alphabet symbols are grouped into PT2 and 048, after applying any required padding rule. The key matrix operates on each three-symbol block. A non-multiple block length is padded with a known symbol such as X; the receiver removes only the padding that was added for that message.

A known-plaintext attack can recover the matrix. If the attacker obtains three linearly independent plaintext vectors and their ciphertext vectors, the equations C=KP can be combined to solve for K modulo 36, provided the plaintext matrix is invertible modulo 36. The simple basis vectors $[1,0,0]^T$, $[0,1,0]^T$ and $[0,0,1]^T$ reveal the columns of K directly.

The demonstration proves that Hill encryption is not secure against sufficient known plaintext. It also shows why invertibility is a mathematical requirement for decryption, not a guarantee of security.

Assignment evidence / study point: Relate this discussion to the healthcare network, identify the assumption being made, and state the limitation of the classical method before drawing a conclusion.

DETAILED NOTES 19: Columnar Transposition: Full Procedure

Columnar transposition rearranges character positions without changing the characters themselves. Choose a keyword or numbered column order. Write the plaintext across rows under the columns. Read the columns in the order determined by the key. Decryption reverses the process by restoring each column to its original position and reading across the rows.

For a four-column key order 3-1-4-2, plaintext is placed in four columns. The columns are then read in ascending key order. If the final row is incomplete, the sender may pad with X or use a recorded length. The receiver must know the padding rule and original message length to recover the exact text.

The cipher does not change letter counts. If the plaintext contains five As, the ciphertext also contains five As. Its histogram and index of coincidence are therefore unchanged.

Columnar transposition can make a message less immediately readable because adjacent plaintext letters are separated. It may disrupt common digrams and word patterns. However, the underlying frequency distribution remains visible. An attacker can try likely column counts, inspect repeated patterns and use language scoring to reconstruct the arrangement.

A columnar transposition is useful for illustrating diffusion through position changes. It is not a strong stand-alone method. The key space for n columns is n! possible column permutations, but the number of practical column counts is limited by message length and the attacker's ability to test arrangements.

In rural clinics, manual columnar transposition may be error-prone. Staff may write a row incorrectly, read columns in the wrong order or forget padding. A simple program can reduce human error, but it does not make the cipher secure.

Assignment evidence / study point: Relate this discussion to the healthcare network, identify the assumption being made, and state the limitation of the classical method before drawing a conclusion.

DETAILED NOTES 20: Key-Space Mathematics: Comparing the Schemes

The Caesar cipher has 26 shifts including the no-change key. Monoalphabetic substitution has 26! possible mappings. Vigenère with a fixed keyword length m has 26^m possible keys, assuming all letters are permitted. A columnar transposition with n columns has n! column orders. A Hill cipher's key space consists of invertible matrices over the chosen modular alphabet; not every matrix is valid.

For a Hill matrix K, the inverse exists modulo m if and only if gcd(det K,m)=1. With modulus 26, the determinant must be relatively prime to 26, meaning it must not be divisible by 2 or 13. With modulus 36, it must be relatively prime to 36, meaning it must not be divisible by 2 or 3.

The condition is important because decryption depends on K^{-1}. A non-invertible key can map two different plaintext blocks to the same ciphertext block, making unique recovery impossible.

A key-space calculation is only one part of security evaluation. A large space may still contain structural weaknesses. Monoalphabetic substitution is the clearest example: 26! is large, but frequency analysis can recover the mapping without searching every key. Vigenère has 26^m keys, but a repeated keyword creates periodic structure. Hill has many possible matrices, but known plaintext can produce linear equations.

A stronger evaluation asks what information the ciphertext leaks, how much ciphertext the attacker has, whether plaintext is predictable, whether keys are reused and whether the attacker can choose or know plaintext. The Day-4 challenge changes several of these assumptions at once.

The report should therefore avoid statements such as “this cipher is secure because its key space is large.” A better statement is: “the key space is this large, but the following attack remains practical under the stated conditions.”

Assignment evidence / study point: Relate this discussion to the healthcare network, identify the assumption being made, and state the limitation of the classical method before drawing a conclusion.

DETAILED NOTES 21: Modular Arithmetic Refresher for Cipher Calculations

Modular arithmetic works with remainders. For example, 29 mod 26 equals 3. In an alphabet of 26 letters, this means moving 29 positions is the same as moving 3 positions. Addition, subtraction and multiplication can all be performed modulo the alphabet size.

For Caesar encryption with P=Z (25) and k=3, C=(25+3) mod 26=2, which corresponds to C. This demonstrates wrap-around from the end of the alphabet to the beginning. Decryption subtracts the key and reduces modulo 26.

For the mod-36 alphabet, a value of 38 becomes 2. The same principle applies to letters and digits. The symbol mapping must be agreed by both sender and receiver; otherwise, the same number may represent different characters.

A modular inverse of a number a modulo m is a value x such that ax is congruent to 1 modulo m. An inverse exists exactly when gcd(a,m)=1. For example, the inverse of 9 modulo 26 is 3 because 27 leaves remainder 1 after division by 26.

The determinant condition for Hill is based on this idea. The matrix inverse formula uses the inverse of the determinant. If the determinant has no modular inverse, the matrix cannot be inverted in the required number system.

Students should show intermediate calculations rather than writing only the final inverse. This makes it possible to check the determinant, greatest common divisor, inverse of determinant, adjugate matrix and final modular reduction.

Assignment evidence / study point: Relate this discussion to the healthcare network, identify the assumption being made, and state the limitation of the classical method before drawing a conclusion.

DETAILED NOTES 22: Cryptanalysis: Brute Force and Frequency Analysis

Brute force tries possible keys until a readable plaintext appears. It is effective against Caesar because there are only 26 shifts. It is not necessary to know the original key in advance. An attacker can produce every candidate and use language knowledge to select the correct result.

Frequency analysis counts how often each letter appears. Natural languages are not uniformly distributed. In English, E, T, A, O, I and N are generally common. A monoalphabetic substitution preserves this frequency pattern under a relabelling. Columnar transposition preserves the exact counts. Therefore, longer ciphertext often gives a clearer frequency estimate.

Frequency analysis is not guaranteed to succeed on very short messages because there may not be enough letters to estimate the distribution. The healthcare scenario includes many messages and predictable formats, so the attacker has more evidence than a single short ciphertext.

A useful demonstration is to compare the frequency histogram of plaintext with the ciphertext histogram. Caesar produces the same counts shifted to different letters. Monoalphabetic substitution produces the same counts under a different label mapping. Columnar transposition produces exactly the same count for every letter. Vigenère may produce a flatter distribution because different key positions use different shifts.

The index of coincidence is calculated as $IC = \Sigma\, f_i(f_i-1) / (N(N-1))$, where f_i is the frequency count of letter i and N is the total number of letters. It measures the chance that two randomly selected letters from the text are identical.

An IC near the expected English value suggests language-like frequency concentration. An IC closer to 1/26 suggests a more uniform distribution. The value is a clue, not a complete proof of security.

Assignment evidence / study point: Relate this discussion to the healthcare network, identify the assumption being made, and state the limitation of the classical method before drawing a conclusion.

DETAILED NOTES 23: Cryptanalysis: Kasiski Examination and IC

Kasiski examination is used against repeating-key polyalphabetic ciphers such as Vigenère. If the same plaintext segment occurs at positions that align with the same key positions, it may produce a repeated ciphertext segment. The distances between repeated ciphertext sequences can be measured. Common factors of those distances may indicate the keyword length.

After estimating the keyword length, split the ciphertext into groups based on character position modulo the key length. Each group has been encrypted using one Caesar shift. Frequency analysis can then estimate each key letter. The attacker combines the recovered shifts to reconstruct the keyword.

The method becomes more reliable when the ciphertext is long and contains repeated language patterns. Many messages encrypted under the same key can provide even more material.

The index of coincidence offers another way to estimate the key period. Calculate IC for the full ciphertext and then calculate IC values for groups formed by candidate periods. If the candidate period is correct, each group may show a more language-like distribution because each group uses one repeated shift.

In the Day-4 challenge, fixed headers such as PATIENT ID and DIAGNOSIS provide known or predictable plaintext. This can expose key characters directly at aligned positions. If many messages reuse the same key, recovered positions from one message can help decrypt others.

Kasiski and IC are statistical techniques. Their success depends on ciphertext length, language, repeated patterns and key behavior. They should be demonstrated with actual test messages rather than presented as guaranteed attacks.

Assignment evidence / study point: Relate this discussion to the healthcare network, identify the assumption being made, and state the limitation of the classical method before drawing a conclusion.

DETAILED NOTES 24: Cryptanalysis: Known-Plaintext Attack on Hill

A known-plaintext attack occurs when an attacker knows one or more plaintext blocks and the corresponding ciphertext blocks. For Hill encryption, $C=KP \bmod m$. If enough independent plaintext vectors are available, the attacker can construct a plaintext matrix P and ciphertext matrix C. When P is invertible modulo m, K can be calculated as $K=CP^{-1} \bmod m$.

The attack depends on the plaintext vectors being independent and the required inverse existing. If the known plaintext matrix is not invertible, the attacker may need additional blocks. This is why the assignment asks for a known-plaintext demonstration rather than only a general statement.

For the 3×3 mod-36 example, basis vectors make the key columns visible. Encrypt $[1,0,0]^T$, $[0,1,0]^T$ and $[0,0,1]^T$. Each output is one column of K. An attacker who obtains these known input-output pairs can reconstruct the matrix.

This attack illustrates a structural weakness rather than a weakness in the number of possible keys. A Hill matrix may have many possible entries, but the encryption rule is linear. The attacker can exploit the equations created by known plaintext.

In a real healthcare network, predictable headers are dangerous because they provide likely plaintext. Repeated field labels, fixed templates and standard phrases may give an attacker enough known material to test the cipher. Removing a fixed header can reduce one source of information, but it does not repair the underlying linear structure.

The correct conclusion is not that Hill is always broken from one known pair. The number and independence of known blocks matter. The assignment should explain the condition under which the key can be recovered.

Assignment evidence / study point: Relate this discussion to the healthcare network, identify the assumption being made, and state the limitation of the classical method before drawing a conclusion.

DETAILED NOTES 25: Cryptanalytic Ranking and Comparative Discussion

The classical methods can be ranked differently depending on the attack model. Against a simple ciphertext-only attacker, Vigenère with a long and varied key may resist basic frequency analysis better than Caesar or monoalphabetic substitution. Hill may also hide single-letter frequencies by mixing blocks. Columnar transposition preserves the histogram, while Caesar and monoalphabetic substitution leak direct frequency structure.

Against a known-plaintext attacker, Hill's linearity is a serious weakness. Against a repeated-key attacker with many messages, Vigenère can be attacked using Kasiski and IC. Caesar remains trivial to brute force regardless of message length. Monoalphabetic substitution remains vulnerable to language statistics even though its key space is large.

A reasonable classroom ranking is Vigenère with a long non-repeating keyword as the strongest of the selected repeating-key classical methods under simple ciphertext-only analysis, followed by Hill for some ciphertext-only conditions, then columnar transposition, monoalphabetic substitution and Caesar. This ranking must be qualified: a reused Vigenère key can be broken, and Hill can fall quickly under known plaintext.

No ranking should imply that any of these methods is suitable for actual patient information. The attacker model matters. The Day-4 insider has many ciphertexts, predictable headers and access to a clinic environment. Under those conditions, the resistance of the original design decreases.

A good conclusion identifies the attack assumptions rather than presenting one universal ranking.

Assignment evidence / study point: Relate this discussion to the healthcare network, identify the assumption being made, and state the limitation of the classical method before drawing a conclusion.

DETAILED NOTES 26: Columnar Transposition of a Lab Table

Consider a lab table with 20 records and four fields: PatientCode, TestCode, Result and Unit. Each row represents one test result. Before transposition, the table should be serialized in a fixed and agreed format, such as one row per line with fields separated by a delimiter. The serialized characters are written into a rectangular grid and read by columns according to the key order.

A row count of 20 ensures that the exercise is larger than a short text example. The table can contain fictional patient codes and values such as HB 13.2 g/dL, WBC 7.1 10^9/L and GLU 96 mg/dL. These values are examples only and are not medical advice.

After encryption, the receiver reverses the column order and reconstructs the serialized table. The receiver then parses the rows and checks that the number of rows and fields matches the expected format.

Transposition changes position but not content. If one result cell changes from 5.6 to 8.6, decryption will reveal the changed value, but the transposition operation itself will not identify the change. If two complete rows are swapped, the table may still contain all original characters and values, only in a different order.

A simple checksum can detect some modifications. Map each character to a numeric value and calculate a sum modulo 97. Store the checksum separately. A changed cell will usually change the sum. However, swapping rows may preserve the sum because addition is independent of order. An attacker can also alter the data and recompute an unkeyed checksum.

The exercise demonstrates the difference between rearrangement, error detection and secure integrity. A classical checksum is not a cryptographic message authentication code.

Assignment evidence / study point: Relate this discussion to the healthcare network, identify the assumption being made, and state the limitation of the classical method before drawing a conclusion.

DETAILED NOTES 27: Checksums: Demonstration and Defeat

A checksum is a value calculated from data so that the receiver can compare the received data with the expected value. For the classroom lab table, assign numeric values to characters and calculate S=(sum of values) mod 97. The receiver recalculates S after decryption. If the values differ, the table may have been modified or corrupted.

For example, if a result field changes from 5.6 to 8.6, the character sequence changes and the checksum will usually change. The receiver can flag the table for review. This is useful for demonstrating basic error detection.

However, a simple sum checksum does not preserve the order of characters. Swapping two rows leaves the sum unchanged. Similarly, changing two values in opposite directions may cancel out under the chosen arithmetic.

An attacker who knows the checksum algorithm can modify the table and calculate a new checksum. The receiver may accept the modified table because the checksum is not secret and does not prove who created it. A checksum can therefore detect accidental errors but is not a strong defence against an active attacker.

A stronger integrity mechanism must bind the data to a secret key or private signing key. It should cover all important fields, including patient identity, order number, dosage, units and freshness information. This is outside the capabilities of the classical ciphers in the assignment.

The correct report conclusion is that the checksum is a teaching tool. It demonstrates detection of some changes and shows exactly how an attacker can defeat it.

Assignment evidence / study point: Relate this discussion to the healthcare network, identify the assumption being made, and state the limitation of the classical method before drawing a conclusion.

DETAILED NOTES 28: Medical Image Encryption: Intensity Shift Modulo 256

An 8-bit grayscale image stores each pixel as an integer from 0 to 255. An intensity-shift method adds a key k to every pixel modulo 256: $E(x,y)=(I(x,y)+k) \bmod 256$. Decryption subtracts the same key: $D(x,y)=(E(x,y)-k) \bmod 256$.

The method is simple and fast. It can be applied to a 256×256 image or a larger image. The key must be an integer from 0 to 255. Since every pixel is shifted by the same amount, the histogram is circularly shifted. The number of pixels at each intensity changes location in the histogram but the overall frequency distribution is preserved.

The method does not hide spatial structure. Neighboring pixels receive the same shift, so local differences and edges remain. A recognizable image may still be visible even though its brightness changes.

Decryption should be verified by comparing the recovered pixel array with the original. A correct test should confirm that every pixel is identical after decrypting the encrypted image. It is not enough to say that the image looks similar; a byte-for-byte or pixel-by-pixel equality test is stronger.

Shannon entropy is calculated from the intensity distribution using $H=-\Sigma p(i)\log_2 p(i)$. Since a uniform one-to-one intensity shift only relabels the 256 intensity values, the histogram probabilities are permuted and the entropy remains unchanged.

The method is not secure encryption. It is a simple modular transformation that preserves too much structure. It is included because the assignment explicitly requests an intensity shift modulo 256.

Assignment evidence / study point: Relate this discussion to the healthcare network, identify the assumption being made, and state the limitation of the classical method before drawing a conclusion.

DETAILED NOTES 29: Medical Image Encryption: Pixel Permutation

Pixel permutation rearranges the positions of pixels according to a secret permutation. If the original image is represented as a one-dimensional array I, encryption places each pixel at a new position $\pi(i)$. Decryption applies the inverse permutation π^{-1} to restore the original order.

The pixel values themselves are not changed. Therefore, the histogram remains exactly the same and Shannon entropy of the pixel-value distribution remains the same. However, neighboring pixels are moved to different positions, so the visible spatial structure is disrupted.

A simple permutation can make an image look scrambled, but it does not change the underlying distribution of pixel values. It also does not provide integrity. If a pixel is changed or a permutation is modified, the system has no built-in way to authenticate the result.

The permutation key must be generated and stored consistently. For an image of width W and height H, the permutation contains W×H positions. A random permutation can be generated for the demonstration, but the receiver must have the exact same permutation or a reproducible key-derived sequence.

A fixed permutation may be vulnerable if the attacker has known images or can compare multiple encrypted images. It is also operationally difficult to manage for large images if the permutation itself must be stored. A deterministic permutation generated from a key is easier to reproduce, but its security depends on the generator and key design.

Compare the method with intensity shift: intensity shift leaves local spatial relationships visible; permutation disrupts adjacency but preserves the histogram exactly. Both are educational transformations, not secure medical image encryption.

Assignment evidence / study point: Relate this discussion to the healthcare network, identify the assumption being made, and state the limitation of the classical method before drawing a conclusion.

DETAILED NOTES 30: Image Histograms and Entropy: How to Interpret Results

A histogram counts how many pixels have each intensity value. For a grayscale image, the horizontal axis contains values 0 to 255 and the vertical axis contains the number of pixels at each intensity. Comparing histograms before and after encryption helps explain what information a method preserves.

For intensity shift modulo 256, the histogram is shifted around the intensity axis. Its shape remains the same, only the intensity labels are moved. For pixel permutation, the histogram is identical because no pixel value changes. Therefore, histogram comparison alone cannot show whether spatial structure has been hidden.

Entropy measures the uncertainty of a pixel value based on its distribution. Higher entropy can indicate a more uniform histogram, but it does not prove that an image is secure. A pixel permutation has exactly the same entropy as the original because it preserves all pixel values.

A complete image experiment should report original entropy, encrypted entropy and decrypted entropy. The decrypted image should match the original exactly. If the encryption operation is reversible and no data is lost, the decrypted entropy should equal the original entropy.

The visual comparison should include the original image, intensity-shift output, permutation output and decrypted outputs. Use a non-sensitive public sample image. Do not alter patient diagnostic images for an academic demonstration without permission.

The interpretation should connect the numbers to the method. If entropy is unchanged, explain that the pixel-value distribution was preserved. If the image still looks recognizable after intensity shift, explain that the method did not disrupt neighboring-pixel relationships.

Assignment evidence / study point: Relate this discussion to the healthcare network, identify the assumption being made, and state the limitation of the classical method before drawing a conclusion.

DETAILED NOTES 31: Steganography: LSB Embedding Procedure

Steganography hides the existence of a message inside another file. In least significant bit (LSB) image embedding, the lowest bit of a pixel channel is replaced by one bit of the secret message. For an 8-bit channel, changing the least significant bit changes the value by at most one.

The sender first converts the short report into bytes. A length field is added so that the receiver knows how many bytes to extract. The bytes are converted into a sequence of bits. The sender walks through selected pixel channels and replaces each channel's least significant bit with the next message bit.

The receiver reads the same channel sequence, collects the least significant bits, reads the length field and reconstructs the original bytes. The report can then be decoded as text.

Capacity depends on image size and the number of channels used. If one bit is used in each grayscale pixel, a W×H image can hold at most W×H bits before overhead. If all three RGB channels are used, the theoretical capacity is 3WH bits. The practical capacity should be lower to reduce visual and statistical distortion.

LSB embedding is fragile. Resizing, filtering, colour conversion or JPEG compression may change least significant bits and destroy the hidden message. PNG is lossless and is therefore more suitable for a basic demonstration. JPEG is lossy and should be tested as an attack on the embedded message.

The report should measure actual capacity and test extraction. It should not claim that a message survived JPEG compression unless the extraction test succeeds.

Assignment evidence / study point: Relate this discussion to the healthcare network, identify the assumption being made, and state the limitation of the classical method before drawing a conclusion.

DETAILED NOTES 32: Steganography: PSNR and Chi-Square Detectability

Peak Signal-to-Noise Ratio (PSNR) compares an original image with a modified image. First calculate mean squared error (MSE), which is the average of the squared differences between corresponding pixel values. For 8-bit images, MAX=255. $PSNR=10\log_{10}(MAX^2/MSE)$. A higher PSNR generally means the two images are numerically more similar.

LSB embedding often produces a high PSNR because each changed channel differs by only one. However, a high PSNR does not prove that the hidden message is undetectable. Statistical tests may detect unusual changes in the least significant bits.

A chi-square approach can compare pairs of values such as 2k and 2k+1. Strong LSB replacement may make paired counts unusually balanced. This can provide evidence of embedding even when the image looks visually unchanged.

The experiment should record image dimensions, number of embedded bytes, theoretical capacity, actual payload size, MSE and PSNR. It should also report whether extraction works before and after JPEG compression. Use the same image for all tests to make comparisons fair.

For medical images, visual similarity is not enough. A one-level pixel change may matter in a region used for diagnosis. A hidden message should not be embedded into the diagnostic original. Use a separate copy, preserve the original and clearly label the modified file.

The conclusion should compare imperceptibility, capacity, robustness and detectability. LSB is a useful classroom technique, but it is not robust and should not be treated as a secure transport method by itself.

Assignment evidence / study point: Relate this discussion to the healthcare network, identify the assumption being made, and state the limitation of the classical method before drawing a conclusion.

DETAILED NOTES 33: Text-Based Steganography in a Text Circular

A simple text-based hiding method can place a short report inside an ordinary-looking circular. One example is an acrostic: the first letter of each sentence spells the hidden message. Another method selects a character at a fixed position after a delimiter. The sender and receiver agree on the extraction rule in advance.

For a demonstration, write a short circular with sentences whose initial letters spell a short word or report code. The receiver extracts the first letter of each sentence and reconstructs the hidden text. The method is easy to explain and requires no image-processing library.

The capacity is very low because the hidden message is limited by the number of sentences or selected positions. The message may also make the circular look unnatural if the writer forces unusual sentence beginnings.

Text-based hiding is fragile. Editing, reformatting, spell-checking or changing sentence order may destroy the message. A reader who suspects an acrostic can inspect the first letters. The method provides no confidentiality: once the extraction rule is known, the hidden message is readable.

The assignment asks for comparison with image LSB. Text hiding has low capacity and can be detected by unusual writing patterns. Image LSB can hold more data but is vulnerable to image transformations and statistical analysis.

In both cases, steganography hides the presence of a message; it does not make the content secret after extraction. The report should therefore be encrypted before it is embedded. Encrypt-then-embed is a better conceptual design than embedding plaintext.

Assignment evidence / study point: Relate this discussion to the healthcare network, identify the assumption being made, and state the limitation of the classical method before drawing a conclusion.

DETAILED NOTES 34: Encrypt-Then-Embed: Medical Imaging Trade-Off

Encrypt-then-embed means that the sender first encrypts the report and then hides the resulting ciphertext inside an image. If the hidden data is discovered, the attacker still needs the decryption key to read it. This combines concealment of existence with confidentiality of content.

The order matters. If plaintext is embedded directly, anyone who discovers the extraction method can read the report. If ciphertext is embedded, discovery does not automatically reveal the report. However, the encryption must be strong enough for the intended threat; classical ciphers in this assignment are not strong enough for real patient information.

The image should be a separate approved copy, not the original diagnostic image. The medical record should retain a clear link between the original image and the modified carrier image so that staff do not confuse the two.

The main trade-off is that embedding modifies image data. A medical scan may contain subtle patterns, and even small pixel changes may affect interpretation. Therefore, the safest decision is not to hide data inside a diagnostic original. If the assignment requires a demonstration, use a public sample image and clearly state that it is a classroom experiment.

The report should also explain robustness. Basic LSB data may be destroyed by JPEG compression, resizing or image enhancement. If the image is transmitted through a system that recompresses it, extraction may fail. A robust steganographic design is more complex and may create additional diagnostic risk.

Conclusion: steganography may be appropriate for controlled demonstrations or non-diagnostic copies, but it is not a substitute for encryption, authentication or secure medical-image handling.

Assignment evidence / study point: Relate this discussion to the healthcare network, identify the assumption being made, and state the limitation of the classical method before drawing a conclusion.

DETAILED NOTES 35: Python Implementation: Program Structure and Modules

The accompanying Python file is organized into small functions. Caesar encryption and decryption are separate functions. Vigenère encryption and decryption use a repeating keyword. Hill-36 uses a 3×3 invertible matrix over the 36-symbol alphabet. Columnar transposition encrypts and decrypts by rearranging columns. LSB embedding and extraction are optional functions that require the Pillow image library.

A modular program is easier to test than one long block of code. Each function has one responsibility and can be tested independently. The round-trip property is the basic correctness check: decrypt(encrypt(message)) must equal the normalized original message.

The program preserves punctuation in Caesar and Vigenère examples. The Hill demonstration extracts the supported alphabet symbols and returns the compact encrypted stream; the caller records the padding count so that decryption can remove only the added padding.

To run the core tests, open a terminal in the folder containing the file and execute python classical_crypto_assignment.py. The program prints PASS when the round-trip assertions succeed. If a test fails, inspect the corresponding encryption and decryption functions and verify that the key and padding values match.

For LSB use, install Pillow with python -m pip install pillow. Then run the embed command with an input PNG, output PNG and short report. Run the extract command on the output PNG. A PNG carrier is used because it preserves pixel values exactly.

The program is educational. It does not include modern authenticated encryption, safe key storage, user authentication or production-grade medical-image processing. Do not use it to protect real patient records.

Assignment evidence / study point: Relate this discussion to the healthcare network, identify the assumption being made, and state the limitation of the classical method before drawing a conclusion.

DETAILED NOTES 36: Python Testing: Correctness, Edge Cases and Reproducibility

A test plan should include ordinary inputs and edge cases. Ordinary inputs include a patient identifier, a diagnosis label and a prescription phrase. Edge cases include an empty message, one-character message, punctuation, lowercase letters, digits, a Hill message whose length is not divisible by three and a transposition message whose length is not divisible by the number of columns.

For every encryption method, test that decryption returns the expected original. If the method normalizes text to uppercase or removes separators, compare against the documented normalized form. A test should not be marked as passed merely because the program does not crash.

The key should also be tested. Caesar should reject or normalize invalid shifts. Vigenère should reject an empty keyword. Hill should use an invertible matrix. Columnar transposition should use a valid non-empty key and a consistent padding rule.

Reproducibility requires documenting the Python version, operating system, machine specification, input sizes, keys, number of repetitions and timing method. Use identical plaintexts for all cipher comparisons. Run each timing test several times and report the median to reduce the effect of temporary system load.

The assignment requests lengths of 100, 1,000, 10,000 and 100,000 characters. Use a fixed deterministic generator or a saved plaintext file so every algorithm receives the same input. Do not compare unrelated messages because different letter distributions may affect the results.

The supplied program performs core round-trip tests and a Caesar timing baseline. Extend the timing loop to include the other implemented algorithms before presenting a full cross-scheme timing graph. State clearly which measurements were actually collected.

Assignment evidence / study point: Relate this discussion to the healthcare network, identify the assumption being made, and state the limitation of the classical method before drawing a conclusion.

DETAILED NOTES 37: Experimental Design: Frequency Histograms and IC

An experiment should compare plaintext and ciphertext generated from the same input. Convert text to uppercase and remove non-alphabet symbols consistently before counting letters. For each scheme, calculate the frequency of A through Z and plot the values as a bar chart. Use the same horizontal axis for all plots.

Caesar changes the labels but not the count distribution. Monoalphabetic substitution also relabels the counts. Columnar transposition preserves the exact histogram. Vigenère may produce a flatter distribution because the same plaintext letter is shifted differently at different key positions. Hill transforms blocks and may also alter single-letter frequencies.

The interpretation should explain the mathematical reason for each observed shape. A chart without explanation does not satisfy the evidence requirement.

Calculate the index of coincidence for the plaintext and each ciphertext. Use the formula $IC=\Sigma f_i(f_i-1)/(N(N-1))$. If N is less than two, the value is undefined or can be reported as zero by the program with a note. Use the same alphabet filtering for every method.

For each cipher, repeat the experiment with multiple message lengths. Short texts produce unstable frequency estimates. Longer texts provide more statistical evidence. For Vigenère, also calculate IC for groups based on candidate key periods.

The results should be interpreted as evidence under the tested messages and keys, not as universal proof. A single IC value cannot establish that a cipher is secure.

Assignment evidence / study point: Relate this discussion to the healthcare network, identify the assumption being made, and state the limitation of the classical method before drawing a conclusion.

DETAILED NOTES 38: Experimental Design: Timing and Attack Success

Measure encryption and decryption time separately. Use a high-resolution timer such as time.perf_counter. For each message size, repeat the operation at least five times and calculate the median. Report the message length on the horizontal axis and time in seconds on the vertical axis. If values differ greatly in scale, use separate charts or a logarithmic axis and explain it.

The expected trend is that simple character-by-character methods require time proportional to message length. Hill performs matrix operations for each block. Columnar transposition writes and reads a grid. Actual timing depends on implementation details and the computer used, so measurements must come from the student's machine.

Do not write invented numbers into the report. A correct experiment with clearly stated limitations is better than fabricated precision.

Attack success rate should be defined before testing. For example, success may mean recovering the exact plaintext or exact key within a fixed time limit. Run multiple trials with different messages and keys at each ciphertext length. Calculate success rate as successful attacks divided by total attempts, multiplied by 100.

For Caesar, brute force should succeed for nearly all readable test messages because the key space is tiny. For monoalphabetic substitution, frequency analysis generally improves as the ciphertext becomes longer. For Vigenère, repeated-key attacks become more reliable when enough text is available. For Hill, known-plaintext success depends on having enough independent blocks.

A graph should include the attack method and assumptions. A success rate without specifying whether the attacker knows plaintext, key length or message format is not meaningful.

Assignment evidence / study point: Relate this discussion to the healthcare network, identify the assumption being made, and state the limitation of the classical method before drawing a conclusion.

DETAILED NOTES 39: Comparative Analysis: Results and Discussion

The results should connect experimental evidence to the earlier mathematical analysis. If Caesar and monoalphabetic substitution show similar frequency concentration to plaintext, that supports the conclusion that they preserve frequency relationships. If transposition has an identical histogram, that supports the claim that it only rearranges positions. If Vigenère appears flatter, that supports the effect of multiple substitution alphabets.

The IC results should be discussed with caution. Caesar and monoalphabetic substitution preserve the frequency counts under relabelling, so IC remains the same. Transposition preserves the exact counts, so IC remains the same. Vigenère's full-text IC may decrease, but the repeated key can be revealed by grouping positions.

Timing results should be used to discuss computational cost, not security by themselves. A cipher that runs quickly is not necessarily secure. A slow cipher is not necessarily resistant to cryptanalysis. For the rural clinic scenario, usability and error rates matter as much as raw processing speed.

Attack success rates should be tied to the attacker model. The Day-4 insider has many messages and predictable headers. That attacker is stronger than someone who sees one short ciphertext. The report should therefore distinguish a short-message demonstration from the repeated-key challenge.

The overall conclusion should be evidence-based: explain what the charts show, what they do not show and how the results influence the final design.

Assignment evidence / study point: Relate this discussion to the healthcare network, identify the assumption being made, and state the limitation of the classical method before drawing a conclusion.

DETAILED NOTES 40: Key Management Across 25 Satellite Clinics

Key management covers key generation, distribution, storage, use, rotation, revocation, backup and destruction. A single shared key across all 25 clinics is easy to explain but creates a large blast radius. If one clinic employee copies the key, messages from many sites may be exposed. The Day-4 challenge specifically demonstrates this weakness.

Per-clinic keys reduce cross-clinic exposure. A key from Clinic 7 should not automatically decrypt messages from Clinic 8. Per-session keys reduce the number of messages encrypted under one key. Key rotation limits how long a compromised key remains useful. Revocation is necessary when a staff member leaves or a device is lost.

These controls create administrative work. Staff need a reliable process that does not require them to memorize complex keys or write them on paper. The system should manage keys through an approved central process.

Key separation by purpose can further reduce risk. Clinical notes, prescriptions and billing messages should not all use the same secret. If one key is compromised, the attacker should not automatically gain access to every message type.

The assignment asks whether key-space mathematics changes when the design is adapted. Per-clinic and per-session keys increase the number of independently managed keys, but they do not change the mathematical weakness of Caesar, Vigenère or Hill. They reduce exposure from key reuse; they do not transform a weak cipher into a strong one.

The final design should state this distinction clearly. More keys can reduce the impact of one compromise, but secure algorithms and authenticated key management are still required in real deployment.

Assignment evidence / study point: Relate this discussion to the healthcare network, identify the assumption being made, and state the limitation of the classical method before drawing a conclusion.

DETAILED NOTES 41: Insider Threat: Revised Threat Model

An insider may be an employee, contractor or other person with legitimate access to a clinic system. The attacker in the Day-4 challenge has copied ciphertexts for months. This changes the problem because the attacker is not limited to a few observations. The attacker has many messages encrypted under the same key and knows that common headers appear at predictable positions.

The earlier assumption that the key remains secret at every clinic is no longer safe. The attacker may have copied the key, observed its use or gained access to a workstation where it is stored. Even if the key itself is not copied, repeated ciphertexts and known plaintext can expose information about it.

An insider may also know the system's message format, staff workflow and timing patterns. This contextual knowledge makes attacks more practical than a purely mathematical brute-force model suggests.

The risk ranking must be updated. Repeated-key cryptanalysis becomes more likely. Unauthorized copying becomes a confirmed behavior rather than a hypothetical threat. Prescription forgery and impersonation remain critical because the insider may know how legitimate messages are structured.

BLP may still prevent the insider from opening records outside the employee's clearance when the application enforces the rules. But if the employee already holds a shared key that decrypts many messages, BLP cannot make that key secret. Access control and key control must be considered separately.

The response should include investigation, key revocation, rotation, audit review, affected-record assessment and clinical validation of prescriptions transmitted during the exposure period.

Assignment evidence / study point: Relate this discussion to the healthcare network, identify the assumption being made, and state the limitation of the classical method before drawing a conclusion.

DETAILED NOTES 42: Predictable Headers and Known-Plaintext Demonstration

A predictable header provides an attacker with likely plaintext. Suppose a message begins with PATIENT ID. If the attacker has the ciphertext at the same positions and knows the encryption method, the attacker can compare the known plaintext with the ciphertext. For Vigenère, the key letter at a position is $K_i=(C_i-P_i) \bmod 26$.

For example, if a known plaintext character is A (0) and the ciphertext character is D (3), the corresponding key character is D (3). If the key repeats, that recovered position may apply to other messages encrypted with the same key. Repeating headers across many messages create multiple opportunities to recover key positions.

This example is simplified, but it demonstrates why known message formats should be included in the threat model.

Removing fixed headers may reduce direct known-plaintext evidence, but it does not solve key reuse. An attacker may still recognize common language, field formats, dates, clinic codes or repeated phrases. Randomized padding can make message lengths and field boundaries less predictable, but padding is not a replacement for a secure cipher.

A practical redesign should avoid one long-lived shared key, use unique per-clinic and per-session keys, rotate and revoke keys, and use modern authenticated encryption in real deployment. Classical methods can be retained only for the academic comparison.

The evidence-based defence should calculate the effect of key reuse and explain why the number of ciphertexts matters. It should not claim that a small formatting change makes the original cipher secure.

Assignment evidence / study point: Relate this discussion to the healthcare network, identify the assumption being made, and state the limitation of the classical method before drawing a conclusion.

DETAILED NOTES 43: BLP During the Insider Challenge

Bell–LaPadula continues to enforce the confidentiality rules it was designed for. A Restricted billing employee should not read a Confidential diagnosis merely because that employee works at a clinic. A Confidential nurse should not read a Secret specialist file under normal rules. The model can still restrict application-level reads and writes if labels and permissions are correctly enforced.

However, BLP does not protect a key from an authorized person who is allowed to use it. If the clinic's encryption key is shared with staff or stored in a location accessible to the insider, the insider may copy ciphertexts and decrypt them outside the BLP-controlled application. The model cannot control information after it has left the enforcement boundary.

This distinction is central to the challenge. Access control governs who can request an operation within the system. Cryptographic key management governs who can decrypt ciphertext. Monitoring and endpoint controls govern what an authorized user can copy or export. No single one of these controls solves every problem.

The final design should keep BLP as one layer, add least-privilege key access, separate keys by clinic and purpose, audit exports, and revoke compromised credentials. Emergency access should remain available but carefully reviewed.

The model should not be described as useless. It remains useful for confidentiality, but it is irrelevant to some attacks once the insider has valid decryption capability or has copied plaintext.

Assignment evidence / study point: Relate this discussion to the healthcare network, identify the assumption being made, and state the limitation of the classical method before drawing a conclusion.

DETAILED NOTES 44: Redesigned Strategy: Retain, Modify or Redesign

The recommended decision is to retain BLP as a confidentiality policy, modify the key-management and message-handling procedures, and redesign the cryptographic protection for real deployment. Retaining the original single-key design is not defensible after hundreds of ciphertexts have been copied. Modifying only the message header is also insufficient.

Each clinic should have an independent identity and key scope. Session keys should be short-lived. Different message purposes should use separate keys. Compromised keys should be revoked and replaced. A central authority should manage key lifecycle without requiring rural staff to manually exchange long secrets.

Randomized padding and variable framing may reduce predictable structure in the classroom design. These are supporting measures only. They do not remove the frequency and linear weaknesses of classical ciphers.

The design must remain usable. Staff should authenticate using a simple approved process, select the patient and record type, and allow the application to apply the correct cryptographic operation automatically. Staff should not choose cipher keys manually for each message. Training should explain how to report suspected compromise and how emergency access works.

The hidden-report requirement should use encrypt-then-embed on a separate non-diagnostic copy. The original scan should remain unchanged. If the image is recompressed, the system should detect that the hidden payload may have been damaged and provide a safe alternative transfer method.

The final design is layered, but its classical components remain an educational model. Real patient records require modern authenticated cryptography.

Assignment evidence / study point: Relate this discussion to the healthcare network, identify the assumption being made, and state the limitation of the classical method before drawing a conclusion.

DETAILED NOTES 45: Final Recommendation: Defence Against Alternatives

The recommended classroom design uses BLP labels for confidentiality, separate clinic/session keys to reduce cross-clinic exposure, key rotation, audit logs, controlled break-glass access and encrypt-then-embed for a separate image copy. Among the classical schemes studied, Vigenère combined with transposition can be discussed as a layered comparison, but it must not be presented as suitable for real records.

Caesar is rejected because brute force is trivial. Monoalphabetic substitution is rejected because frequency analysis defeats its large key space. Columnar transposition is rejected as a stand-alone cipher because it preserves the histogram. Hill is rejected for real use because it is linear and vulnerable to known plaintext. Repeating-key Vigenère is rejected for real use because repeated keys and known headers expose periodic structure.

The strongest reason for redesign is the insider challenge. Hundreds of ciphertexts under the same key give the attacker more evidence. Predictable headers create known-plaintext material. A shared key increases the number of clinics affected by one compromise. Per-clinic and per-session keys reduce the blast radius but do not fix weak cipher algorithms.

The final recommendation should be honest about residual risks. An authorized insider may still misuse access. A compromised endpoint may expose plaintext. Emergency access may be abused. A network outage may prevent access to specialist advice. These risks require monitoring, incident response, clinical governance and recovery plans.

The final answer is therefore not that classical ciphers solve the healthcare problem. The answer is that classical ciphers demonstrate important principles and limitations, while real patient data requires modern cryptographic protection.

Assignment evidence / study point: Relate this discussion to the healthcare network, identify the assumption being made, and state the limitation of the classical method before drawing a conclusion.

DETAILED NOTES 46: Viva Preparation: Key Concepts and Short Answers

1. What is plaintext? Plaintext is the original readable information before encryption.
2. What is ciphertext? Ciphertext is the transformed output produced by encryption.
3. What is a key? A key is the secret value that controls the encryption or decryption transformation.
4. What is a substitution cipher? It replaces plaintext symbols with other symbols.
5. What is a transposition cipher? It rearranges the positions of plaintext symbols.
6. Why is Caesar weak? It has only 26 possible shifts and preserves frequency patterns.
7. Why does monoalphabetic substitution fail? It preserves frequency and repeated-pattern information.
8. What is a polyalphabetic cipher? It uses multiple substitution alphabets, such as Vigenère.
9. What is the Hill cipher? It encrypts blocks of letters using matrix multiplication modulo an alphabet size.
10. When is a Hill key invertible? When gcd(det K,m)=1.
11. What is Kasiski examination? A method that estimates the repeated keyword length in Vigenère ciphertext.
12. What is index of coincidence? A measure of how likely two randomly selected letters are to be equal.
13. What is BLP? A confidentiality model using security levels and information-flow rules.
14. What is no read up? A subject cannot read an object at a higher level.
15. What is no write down? A subject cannot write information to a lower-level object.
16. What is steganography? Hiding the existence of a message inside another carrier.
17. Why is LSB fragile? Image transformations and JPEG compression may change the least significant bits.
18. Why are classical ciphers unsuitable for patient data? They lack modern authenticated security and are vulnerable to known attacks.

Assignment evidence / study point: Relate this discussion to the healthcare network, identify the assumption being made, and state the limitation of the classical method before drawing a conclusion.

DETAILED NOTES 47: Viva Preparation: Scenario-Based Questions

Question: Why is prescription modification more serious than ordinary data disclosure? Answer: A changed dosage or medicine may directly harm a patient, whereas disclosure primarily affects privacy, although both are serious.

Question: Can a correctly decrypted message prove the sender is a doctor? Answer: No. A shared-key holder can create ciphertext. Decryption only shows that the message is compatible with the key and algorithm.

Question: Why does the insider challenge change the risk? Answer: The attacker has many ciphertexts under the same key and knows predictable headers, making known-plaintext and statistical attacks more effective.

Question: Does BLP stop a staff member from copying a key? Answer: Not necessarily. BLP controls information-flow access inside its enforcement boundary; key storage and endpoint controls are separate.

Question: Why use per-clinic keys? Answer: A compromised key should not expose messages from every clinic. Per-clinic keys reduce the blast radius.

Question: Does a large key space prove a cipher is secure? Answer: No. Monoalphabetic substitution has 26! keys but preserves frequency patterns and can be broken by analysis.

Question: Why should the original medical image remain unchanged? Answer: Even small pixel changes may affect diagnostic interpretation. A separate copy should be used for a steganography demonstration.

Question: What is the final real-world recommendation? Answer: Use modern authenticated encryption, secure key management, access controls, audit logging, backups and emergency procedures. Classical ciphers are only for academic study.

Assignment evidence / study point: Relate this discussion to the healthcare network, identify the assumption being made, and state the limitation of the classical method before drawing a conclusion.

DETAILED NOTES 48: Submission Checklist and Evidence Guide

Before submission, check that the report contains all 12 tasks in the order shown in the assignment PDF. Task 1 should include at least six attacks and a justified risk ranking. Task 2 should map goals to the three functions. Task 3 should include security levels, at least five access decisions and a critical evaluation. Task 4 should show at least three distinct schemes with formulas and worked examples. Task 5 should derive key spaces and show a Hill inverse.

Task 6 should explain suitable cryptanalytic attacks and rank resistance. Task 7 should discuss transposition and product ciphers. Task 8 should compare steganography with encryption. Task 9 should include working code and reproducible tests. Task 10 should include actual tables and graphs. Task 11 should defend the final design. Task 12 should address the insider challenge in four parts.

Attach screenshots of the Python program running successfully. Include screenshots of at least one encryption/decryption test, one Hill calculation, one image LSB extraction test if completed, and any charts generated from actual measurements. Label every figure with a number and a descriptive caption.

Do not fabricate experimental measurements. If a requested experiment cannot be completed, state the limitation and provide the procedure and expected interpretation. The assignment expects evidence, so actual screenshots and measured data will improve the submission.

Finally, check spelling, page numbering, headings, table alignment, name, register number and submission date. Ensure that the conclusion states clearly that classical ciphers are not suitable for real patient records.

Assignment evidence / study point: Relate this discussion to the healthcare network, identify the assumption being made, and state the limitation of the classical method before drawing a conclusion.

DETAILED NOTES 49: References and Further Reading

The primary source for the assignment structure, scenario, task requirements and mark distribution is the CO1 Assignment 1 PDF supplied with the question paper. The report follows its 12-task organization and uses the healthcare network and Day-4 challenge described there.

Additional learning references consulted for classical cipher concepts include the chapter “Classical Ciphers and Their Cryptanalysis” in Cryptography, Information Theory, and Error-Correction (Wiley, 2021), which discusses historical ciphers, frequency analysis and attacks on Vigenère. A University of Rhode Island teaching page on the Hill cipher explains block encryption and the requirement that the key matrix be invertible modulo 26. These sources support the educational discussion, not the claim that classical ciphers are suitable for modern patient data.

Other useful references include William Stallings, Cryptography and Network Security: Principles and Practice, for classical encryption techniques and security services; and Bell and LaPadula's work on secure computer systems for the confidentiality model. The references should be checked against the edition and format available through the college library before final submission.

Web references:

- Wiley chapter, “Classical Ciphers and Their Cryptanalysis”: https://doi.org/10.1002/9781119582397.ch2
- University of Rhode Island, “Hill Cipher”: https://www.cs.uri.edu/cryptography/classical-ciphers/hill/article.html
- IIT Bombay lecture notes, “Introduction and Classical Cryptography”: https://www.ee.iitb.ac.in/~sarva/courses/EE720/2024/slides/introduction.html

These references are supplementary. The supplied assignment PDF remains the controlling source for the required tasks and marking scheme.

Assignment evidence / study point: Relate this discussion to the healthcare network, identify the assumption being made, and state the limitation of the classical method before drawing a conclusion.

Expanded Assignment Index

- Front matter and scenario overview
- Tasks 1–3: risk assessment, security goals and Bell–LaPadula
- Tasks 4–7: cipher design, key-space mathematics, cryptanalysis and transposition
- Tasks 8–10: steganography, implementation, testing and experimental evidence
- Tasks 11–12: final recommendation and Day-4 insider challenge
- Detailed notes 01–42: expanded explanations, worked reasoning, operational considerations and viva preparation
- Appendix: Python source file supplied separately